

# Allocation problem in cross-platform ride-hail integration

Ruijie Li <sup>a,b,c</sup>, Yang Liu <sup>c,d</sup>, Xiaobo Liu <sup>a,b</sup>, Yu (Marco) Nie <sup>e,\*</sup>

<sup>a</sup> School of Transportation and Logistics, Southwest Jiaotong University, Chengdu, China

<sup>b</sup> National Engineering Laboratory of Integrated Transportation Big Data Application Technology, Southwest Jiaotong University, Chengdu 611756, China

<sup>c</sup> Department of Civil and Environmental Engineering, National University of Singapore, 117576, Singapore

<sup>d</sup> Department of Industrial Systems Engineering and Management, National University of Singapore, 117576, Singapore

<sup>e</sup> Department of Civil and Environmental Engineering, Northwestern University, IL 60208, USA

## ARTICLE INFO

### Keywords:

Stackelberg game

Ride-hail

Artificial scarcity

Third-party integration

Bilevel pricing problem

## ABSTRACT

We consider a ride-hail system in which a third-party integrator receives ride requests and allocates them to ride service platforms. The ride allocation problem (RAP) is modeled as a Stackelberg game. The integrator, as the leader, chooses the allocation that maximizes its profit, by pricing the rides such that no platform (i.e., follower) can find a more profitable allocation. In pursuit of self-interest, the integrator may refuse to match as many rides as the platforms are willing to serve, thereby injecting an artificial scarcity into the system. To protect the platforms from over exploitation, an exogenous reserve price is introduced to bound their per capita profit from below. We formulate RAP as a bilevel pricing problem, and convert it to a single-level problem by dualizing the lower level. When artificial scarcity is eliminated and all reserve prices are set to zero, we prove the single-level problem can be turned into a mixed integer-linear program that equals its linear relaxation, thus becoming polynomially solvable. Moreover, this version of RAP is shown to be related to cooperative assignment games. Numerical experiments confirm that artificial scarcity negatively affects matching productivity and social welfare. The integrator is favored to take most profits, and leveraging artificial scarcity strengthens its dominance. Moreover, the tighter the supply, the more the integrator benefit from artificial scarcity. The reserve price helps redistribute benefits from the integrator to the platforms. However, demanding an excessively large reserve price may depress the platforms' profits, while undermining system efficiency.

## 1. Introduction

In the past decade or so, smartphone-based ride-hail has grown from serving a niche market to a ubiquitous, if not indispensable, mode of transportation around the world (Cramer and Krueger, 2016; Nie, 2017; Zhang and Nie, 2021). As the market begins to saturate,<sup>1</sup> the ride-hail industry is eagerly looking for innovations to drive growth. In China, where the industry is still highly fragmented<sup>2</sup> and the competition for dominance remains fierce, a recent innovation was third-party integration.

\* Corresponding author.

E-mail address: [y-nie@northwestern.edu](mailto:y-nie@northwestern.edu) (Y.(M. Nie).

<sup>1</sup> A recent Chinese government report shows the number of ride-hail users in China had peaked in December of 2018, see [https://www.cac.gov.cn/2021-02/03/c\\_1613923423079314.htm](https://www.cac.gov.cn/2021-02/03/c_1613923423079314.htm).

<sup>2</sup> According to a recent survey <https://ecoapp.qianzhan.com/details/220525-fd18bcc5.html>, there are over 250 ride-hail operators in China, of which 18 served more than 300,000 rides in March of 2022.

A third-party integrator is usually a location-based service provider that is positioned to sell transportation as an add-on service to its users. For example, a popular map service like Baidu Map and AutoNavi map may handle millions of location requests each hour. As many of these requests are made by customers with the intention to visit a business physically, the map service can seize upon the opportunity to offer a ride that is not of their own making, but provided through a partnership with ride-hail service platforms (e.g., Didi Chuxing and Caocao Chuxing in China). For the integrator, the appeal of such an add-on service is obvious. It broadens the portfolio of their services — which helps attract more users — and opens up a revenue stream from the portion of the profit they are entitled to retain. On the other hand, the partnership benefits ride-hail platforms too, as the integrator can provide them with a much larger customer base than what they would otherwise have access to — this is especially true for smaller platforms. The integrator-led ride-hail business model has grown substantially in China, even though it is less developed elsewhere. AutoNavi map reportedly processes nearly 30% of China's ride-hail trips nowadays, or about 8 million orders a day,<sup>3</sup> and works with more than 160 platforms.<sup>4</sup> Because the integrator retains multiple platforms, they face the task of allocating ride requests, in real-time, among its partners. Intuitively, the integrator, acting as an intermediary in a two-sided market, must make allocation in such a way that not only (i) maximizes its own profit; but also (ii) serves as many riders as possible; and (iii) be fair to every partner to maintain a stable coalition. It is the formulation and the solution of this allocation problem that motivates our study.

We propose to formulate the ride allocation problem (RAP) in cross-platform integration as a Stackelberg, or leader-follower, game, where followers (platforms) prefer a set of rides that maximize their own profit and the leader (the integrator) prices the rides to influence the preference of each follower. To protect them from being overly exploited, the platforms are given<sup>5</sup> a minimum profit per ride, referred to as their *reserved price*. The central question at hand is the following: at any given moment, for a given set of known rides and a set of platforms, how to set the price of each ride such that the allocation satisfies not only the leader (objective (i) above), but also riders (objective (ii)) and platforms (objective (iii)). A solution to the Stackelberg game naturally meets objectives (i) and (iii), because by definition neither the leader nor the followers would have incentives to deviate from their respective strategies — price selection for the integrator and ride selection for the platforms — at that solution. However, simply solving the Stackelberg game might not satisfy riders. At issue here is the conflict between the integrator's monopoly power over their *own* platforms and the reality that they must compete with other integrators and platforms. The monopoly power allows the integrator to squeeze as much profit as possible, even when doing so means intentionally “pricing out” requests that some platforms would be willing to serve. In economics, this phenomenon is known as *artificial scarcity* (Bronfenbrenner, 1962). Since any given integrator's monopoly power is limited in scope, profit-maximization through artificial scarcity is myopic, in the sense that, in the long run, it can lower the overall productivity of the integrator, thus undermining the appeal to riders. Therefore, we shall assume that a rational integrator prefers eliminating artificial scarcity in favor of overall customer satisfaction and competitiveness. This requirement may be incorporated into the Stackelberg game as a separate constraint.

A Stackelberg game is typically formulated as a bilevel program (Kleinert et al., 2021). Because of the hierarchy, bilevel programs are difficult to analyze and solve in general. On the one hand, the upper level is a pricing problem with a bi-linear objective function, integer decision variables, and a hard, nonlinear constraint needed for eliminating artificial scarcity. On the other hand, the lower level consists of a series of bipartite matching problems, one for each platform. In this study, we address the above challenge in two steps. Firstly, we develop a transformation of the original RAP by introducing a virtual platform that can serve any ride request while exactly breaking even. We then prove that, for the transformed RAP, artificial scarcity is naturally eliminated, without imposing the hard nonlinear constraint. Secondly, we reformulate the bilevel program as a single-level problem based on duality. The problem can be transformed to a mixed-integer linear program (MILP) despite the application of the strong-duality condition introduces nonlinearity. When all reserve prices are set to zero, we further prove the MILP can be reduced to a linear program (LP), thus becoming polynomially solvable. Indeed, we shall see that this particular RAP is related to the cooperative assignment game of Shapley and Shubik (1969), which maximizes both the total profit of all stakeholders (sometimes known as social welfare) and the profit share of the buyer (the integrator in our problem). Thus, removing artificial scarcity and reserve price is a sufficient condition to induce full cooperation between the integrator and the platforms. To the best of our knowledge, this is a new result.

For the remainder, Section 2 reviews related studies. Section 3 sets up the model and presents the Stackelberg game formulation. Section 4 reduces the Stackelberg game to an MILP and elaborates the connection between it and the cooperative assignment game. Section 5 provides and discusses the results of numerical experiments and Section 6 concludes the study with a summary of findings and ideas for future research.

## 2. Related studies

Our work is built on ridesharing in general and bipartite matching in particular. Both are actively researched areas with a rapidly growing literature. We shall not attempt a comprehensive review of these topics here. The reader interested in reading one may consult (Agatz et al., 2012; Furuhashi et al., 2013; Wang and Yang, 2019; Mourad et al., 2019; Tafreshian et al., 2020; Mitropoulos et al., 2021; Qin et al., 2021). Instead, our focus is on third-party integration in mobility service markets, as well as bilevel pricing problems, to which RAP belongs.

<sup>3</sup> See <https://baijiahao.baidu.com/s?id=1758610021187784046&wfr=spider&for=pc>.

<sup>4</sup> See <https://baijiahao.baidu.com/s?id=1746541177036242231&wfr=spider&for=pc>.

<sup>5</sup> In other words, this is not modeled as a decision of the platform.

### 2.1. Third-party integration

Studies of third-party integration are new and relatively scarce. Using a macroscopic ride-hail mode similar to [Castillo et al. \(2017\)](#), [Zhou et al. \(2022\)](#) analyzed the impact of third-party integration on the performance of a ride-hail market served by multiple ride-hail platforms, which are either competing (Nash equilibrium) or cooperating (social optimum) with each other. They found integration increases trip production and social welfare in both cases, but not necessarily the profit. These phenomena are explained by three mechanisms embedded in the model with integration: an elevated market sickness, a reduction in matching friction, and a higher competitive pressure. [Zhou et al. \(2022\)](#) assumed the integrator has the power to directly match ride requests with any of the vacant vehicles operated by its platforms. This assumption was relaxed in [Bao et al. \(2023\)](#), which instead considers two arrangements. In the first, the integrator simply sends a ride request to all platforms and assigns it to the platform that offers the minimum wait time (based on the platform's own supply). Their second arrangement allows the passengers to select a subset of platforms, from which the integrator then picks the best (again, measured by the wait time) for them. They confirmed the main finding by [Zhou et al. \(2022\)](#) – that integration boosts trip production and social welfare – stands even when the integrator has no direct control of the platforms' fleets. They also found “high-end” platforms (i.e. those charging a higher fare, presumably in exchange for better user experience) are better off in the first arrangement, and the low-end platforms benefit more in the second. [Dong et al. \(2024\)](#) model the interaction between an on-line service platform (OSP) and its teams of workers as a Stackelberg game. Specifically, the OSP is the leader setting rewards and intra-team revenue-sharing rules, and the teams (followers) compete with each other to maximize the expected utility. They found team contest always benefits workers and teamwork (intra-team coordination) mitigates over-competition among workers. Although this problem appears to have little to do with third-party integration, we note the teams and OSP in their game can be viewed as mathematical equivalents of platforms and integrator, respectively.

Built on the centralized ridesharing model of [Alonso-Mora et al. \(2017\)](#), [Pandey et al. \(2019\)](#) also studied cooperation and competition between platforms. In their competitive model, the free competition for ride requests by platforms leads the system to a Nash equilibrium. Their cooperative model, on the other hand, solves a system optimal bipartite matching problem by a distributed auction algorithm, which waives the need for full information about matching costs usually required in bipartite matching. “Cooperation” in this case is actually achieved through a multi-round competitive bidding process coordinated by a broker (integrator). They found, as expected, the cooperative model outperforms the competitive model in terms of the quality of the ridesharing service. Conceptually, the cooperative model of [Pandey et al. \(2019\)](#) is similar to the first arrangement in [Bao et al. \(2023\)](#), although they represent matching decisions at completely different levels of aggregation.

Third-party integration has also been considered in the context of Mobility-as-a-Service (MaaS), whereby the integrator, or the MaaS platform, is an intermediary sitting between travelers and transportation service providers (TSPs). [Xi et al. \(2022\)](#) modeled the interaction between travelers and TSPs as a single-leader multi-follower game, in which both travelers and TSP seek to optimize a private objective function by adjusting their participation level in MaaS, while the integrator seems to optimize its profit. While they hence recognized the hierarchy of the game, they did not explicitly consider matching, hence significantly simplifying the “physics” of resource allocation. Another setup assumes TSPs operate certain routes in a network, to which travelers are assigned. However, the “MaaS integration” spoken of in the latter case usually does not have a bi-level hierarchy. For example, [Rasulkhani and Chow \(2019\)](#) formulated such a single-level assignment game and analyzed the properties of stable pricing. [Pantelidis et al. \(2020\)](#) extended the game to allow travelers to make multi-modal, multi-operators trips. [Ding et al. \(2023\)](#) studied a similar assignment game, though their focus was to ensure allocation and pricing are incentive compatible and rational, and to develop off-line and online algorithms.

### 2.2. Bilevel pricing problem

Bilevel optimization problems have been extensively studied since 1970s ([Colson et al., 2007](#); [Kleinert et al., 2021](#)). Under this general label is a special category, known as bilevel pricing problem ([Bialas and Karwan, 1984](#); [Brotcorne et al., 2001](#); [Zugno et al., 2013](#); [Grimm et al., 2021](#)), which has only linear constraints and bilinear objectives. Bi-linearity means the lower (upper) level problem is reduced to a linear one when the upper (lower) level decisions are fixed. A classical example is the road pricing problem ([Labbé et al., 1998](#)), in which the toll authority (leader) levies taxes on a set of road segments while travelers (followers) optimize their own travel cost (the sum of toll and travel time). The followers' objective contains the product of a binary variable (the lower level decision indicating whether a toll road is used) and the toll level (an upper level variable), a bilinear term that becomes linear once the toll level is fixed. We shall see in Section 3 RAP is also a bilevel pricing problem. Thus, the rest of this section will focus on the analysis of this problem.

Taking advantage of bi-linearity, the standard approach to a bilevel pricing problem is reformulating it as a single-level optimization problem. This is usually accomplished in two ways. The first is to write down the KKT conditions of the lower level problem and add them into the upper level as constraints ([Zugno et al., 2013](#); [Dempe, 2019](#)). The other method first obtains the dual of the lower-level problem, and then augments the constraints of the upper level with those from the dual, along with the strong duality condition ([Grimm et al., 2021](#); [Kleinert and Schmidt, 2021](#)). In both methods, however, the resulting single level problem contains nonlinear equality constraints, which come from the complementarity conditions in the KKT method, and from the strong duality in the dual method. In the literature, these nonlinear constraints are usually linearized by introducing binary variables (sometimes known as the big-M method, see [Kleinert et al., 2021](#)). This produces a mixed integer linear or nonlinear (depending on the objective of the upper level) problem (MILP or MINLP) that can be solved by off-the-shelf solvers (see e.g.,

Farvaresh and Sepehri, 2011). Of course, neither is solvable in polynomial time, and thus one cannot hope to solve large instances in a reasonable amount of time even with most sophisticated solves available. Specialized algorithms that bypass standard linearization by big-M (e.g., Bender's decomposition, see Fontaine and Minner, 2014) may improve computational efficiency under favorable conditions, though their promise is understandably limited by the combinatorial nature of the problem (Kleinert et al., 2021).

### 2.3. Summary

Our reading of literature indicates the RAP has not been studied before. Except for Pandey et al. (2019), the studies reviewed in Section 2.1 either employ aggregate models that rule out individual matching details central to RAP or do not consider a hierarchical game. The cooperative model in Pandey et al. (2019) treats the broker as a disinterested facilitator aiming to maximize system efficiency. This is in sharp contrast with RAP, in which the integrator is a profit-maximizing firm that manipulates self-interested platforms through pricing.

The analysis of the RAP presented in this paper follows the standard approach in the bi-level pricing literature. The novelty lies in three aspects: (i) by introducing a virtual platform, artificial scarcity is eliminated without imposing any hard constraint; (ii) an MILP formulation of RAP is obtained by taking advantage of the RAP's special structure; and (iii) we prove this MILP can be further reduced to a linear program when all reserve prices are zero.

We close by noting the RAP with zero reserve price is related to cooperative assignment games, which can be traced back to the study of the marriage problem by Gale and Shapley (1962). By introducing the concept of stable matching, Gale and Shapley (1962) laid the foundation for studying one-to-one assignment games. Shapley and Shubik (1971) examined the properties of the core of an assignment game, which defines a stable payoff allocation from which any subgroup of players has no incentive to deviate. They also formulated the assignment game as a linear program (LP) and showed the core is a non-empty complete lattice of the program. Demange (1982) further proved that each buyer (seller) attains their marginal contribution in buyer-(seller)-optimal core. Sotomayor (1999) extended the assignment game to allow players to form partnerships, and showed the game admits a unique optimal stable payoff for each side of the market. We shall see that, when artificial scarcity is eliminated and all reserve prices are zero, the solution to the RAP is the buyer-optimal core of a collaborative assignment game similar to that studied in Sotomayor (1999). This relationship between the core of an assignment game and the artificial scarcity introduced by a leader-follower hierarchy has not been previously noticed in the literature, to the best of our knowledge.

## 3. Problem statement

### 3.1. Model settings

Consider an integrator that receives a set of  $\mathcal{R}$  ride requests (referred to as rides hereafter) within a given decision episode (say 30 s). Upon receiving a ride request  $r \in \mathcal{R}$ , it first calculates the fare of the ride based on the time of day and origin/destination (e.g., the fare may be determined based on the cost per unit travel distance and the cost per unit travel time). The fare, denoted as  $\tilde{q}_r$ , is what the rider would see and pay on their mobile device. To serve the rides, the integrator procures services from a set of ride-hail platforms  $\mathcal{P} = \{1, 2, \dots, P\}$ . Suppose for a given episode, each platform  $p \in \mathcal{P}$  has a fleet of available vehicles  $\mathcal{V}_p$ . Let  $\mathcal{V} = \cup_{p \in \mathcal{P}} \mathcal{V}_p$  denote the set of all vehicles and  $\delta_{vp} = 1$  if vehicle  $v$  belongs to platform  $p$ , otherwise  $\delta_{vp} = 0$ .

The integrator is essentially a middleman, or a market maker. To riders, it acts as a seller of transportation service, but to platforms, it is the buyer of their services. Accordingly, the fare  $\tilde{q}_r$  is the seller's price (i.e., the price paid by rider  $r$ ) and we define  $q_r$  as the buyer price, which is the price the integrator pays a platform to procure its service. The integrator makes money from the spread  $\tilde{q}_r - q_r, \forall r \in \mathcal{R}$ . It is worth emphasizing that the platforms can see  $q_r$ , but not  $\tilde{q}_r$ .

We assume the ride-hail platforms enter into a contract with the integrator, characterized by the following terms: (1) the integrator dictates the buyer price  $q_r$  and allocates rides to the platforms such that, given the posted price, each rider-hail platform always receives a set of rides that maximizes its own profit; (2) each ride-hail platform  $p$  discloses (i) its fleet information and the cost required to deliver ride  $r$  by any vehicle  $v \in \mathcal{V}_p$ , denoted as  $c_{vr}$ <sup>6</sup>; and (ii) a minimum profit expected for each ride served, referred to as the reserve price and denoted as  $\zeta_p$ . The reserve price of a platform may depend on such factors as its average fleet size, service quality, and market share. Since these factors are invariant in the short run, we assume  $\zeta_p$  is set in the contract, thus treated as exogenous to the integrator's ride allocation problem. The reserve price is further assumed to be proportional to the delivery cost  $c_{vr}$ . Specifically, letting  $\zeta_{vr}$  denote the reserve profit realized when vehicle  $v$  is matched to ride  $r$ , we have

$$\zeta_{vr} = \sum_{p \in \mathcal{P}} \delta_{vp} \zeta_p c_{vr}. \quad (1)$$

The integrator's ride allocation problem is formulated as a Stackelberg (or leader-follower) game, with the integrator assuming the role of the leader and the ride-hail platforms that of the follower, see Fig. 1 for an illustration. All players aim to maximize their own profits. The integrator's decision variable is the buyer price  $q_r$ , according to which the platforms determine ride allocation. We use  $x_{vr}$  to denote the allocation decision:  $x_{vr} = 1$  if the matching pair  $(v, r)$  is matched (i.e., ride  $r$  is allocated to vehicle  $v$ ); and

<sup>6</sup> In addition to the cost of transporting the rider from their origin to their destination, the delivery cost may also be influenced by the pickup cost and the expected return around the drop-off location (which depends on the potential demand). Be complex as it may, estimating the delivery cost  $c_{vr}$  is exogenous to the ride allocation problem, and will be treated as such in this study.

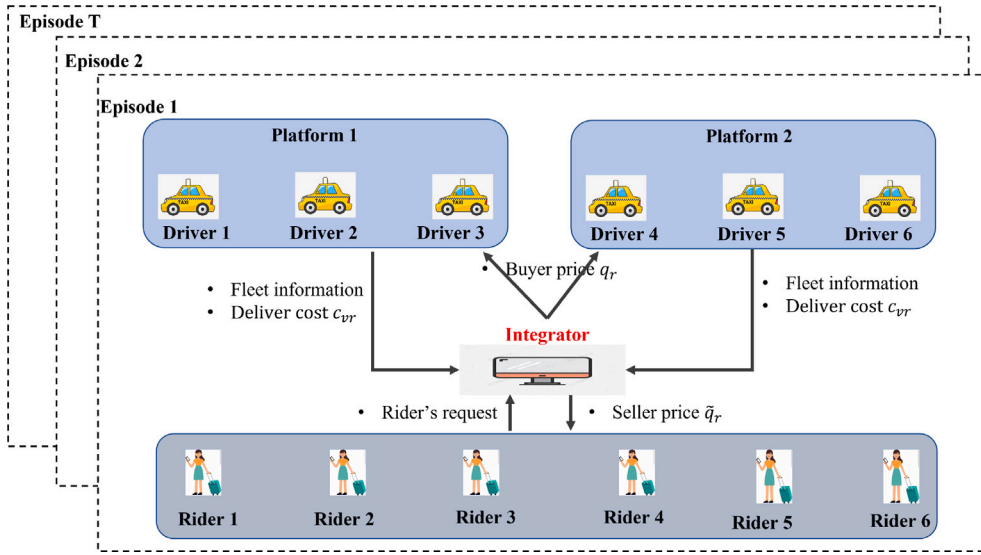


Fig. 1. Structure of the Stackelberg game in which the integrator is the leader, and the ride-hail platform are the followers.

0 otherwise. For convenience, we define the vectors of main variables using boldface symbols. Specifically,  $\mathbf{q} = \{q_r, r \in \mathcal{R}\}$ ,  $\tilde{\mathbf{q}} = \{\tilde{q}_r, r \in \mathcal{R}\}$ ,  $\mathbf{x}_p = \{x_{vr}, r \in \mathcal{R}, v \in \mathcal{V}_p\}$ ,  $\mathbf{x} = \{x_p, \forall p \in \mathcal{P}\}$ ,  $\mathbf{c} = \{c_{vr}, r \in \mathcal{R}, v \in \mathcal{V}\}$ . It is worth emphasizing that, ultimately, actual ride allocation is controlled by the integrator, who is poised to anticipate and manipulate the platforms' preferences (as expressed in their profit-maximization problem) through pricing. Thus, even each platform is self-interested, collectively they are incentivized to behave in a way deemed "optimal" by the integrator (or the system as a whole).

The integrator's ride allocation problem (RAP) is to assign a set of rides  $\mathcal{R}$  to a set of platforms  $\mathcal{P}$  such that the integrator's profit, i.e., the total fare collected from customers less the total payment to platforms, is maximized subject to the following two conditions. First, each platform must agree the rides it receives from the integrator are the best possible, in the sense that swapping any of its current rides with another ride cannot increase its profit. In the literature, this is known as the stable matching condition (Shapley and Shubik, 1969). Second, no artificial scarcity exists, i.e., no rides should be left unserved when there exists at least one platform willing to serve it. Artificial scarcity arises when the integrator's interest is at odds with the welfare of the system over which it exercises a monopoly power. Although the integrator fully controls the allocation of rides, it still has to compete with other integrators and independent platforms. Thus, we posit that the integrator would deem artificial scarcity harmful to its long-term competitiveness, even though it might boost short-term profitability.

### 3.2. Artificial scarcity

We next characterize two types of artificial scarcity. The first type, called strong artificial scarcity, arises when neither ride  $r$  nor vehicle  $v$  is matched, but the profit of the matching pair  $(v, r)$ , if realized, would exceed the reserve profit ( $\zeta_{vr} \leq \tilde{q}_r - c_{vr}$ ). We formally define it as follows.

**Definition 1 (Strong Artificial Scarcity).** A given allocation and price vector  $[\mathbf{x}, \mathbf{q}]$  introduces strong artificial scarcity into the RAP if and only if there exists at least one matching pair  $(v, r)$  that satisfies Condition (2).

$$\sum_{v' \in \mathcal{V}} x_{v'r} = 0, \quad \text{and} \quad (2a)$$

$$\sum_{r' \in \mathcal{R}} x_{vr'} = 0, \quad \text{and} \quad (2b)$$

$$c_{vr} + \zeta_{vr} \leq \tilde{q}_r, \quad \text{and} \quad (2c)$$

$$q_r < c_{vr} + \zeta_{vr}. \quad (2d)$$

Note that Eqs. (2a) and (2b) dictate, respectively, that vehicle  $v$  is unmatched and rider  $r$  left unserved at the current allocation. Eq. (2c) states the potential profit of the unrealized matching pair  $(v, r)$  is in fact larger than the reserve profit. Eq. (2d) ensures the pair  $(v, r)$  be unrealized because the integrator prices it below the sum of the platform's cost and reserve profit, despite the fare is enough to cover it.

To eliminate strong artificial scarcity, one may impose, for all pairs  $(v, r)$  that satisfy  $c_{vr} + \zeta_{vr} \leq \tilde{q}_r$ , the following constraint on the allocation and price decisions,

$$\left(1 - \sum_{v' \in \mathcal{V}} x_{v'r}\right) \left(1 - \sum_{r' \in \mathcal{R}} x_{vr'}\right) (q_r - c_{vr} - \zeta_{vr}) \geq 0, \quad (3)$$

where  $c_{vr}$ ,  $\zeta_{vr}$ , and  $\tilde{q}_r$  are fixed parameters. First note that a pair  $(v, r)$  such that  $c_{vr} + \zeta_{vr} > \tilde{q}_r$  cannot cause strong artificial scarcity per Condition (2). Otherwise, satisfying Condition (3) rules out Condition (2), because the only possible way to violate Condition (3) is  $\sum_{v' \in \mathcal{V}} x_{v'r} = 0$ ,  $\sum_{r' \in \mathcal{R}} x_{vr'} = 0$ , and  $q_r - c_{vr} < \zeta_{vr}$ . In other words, if Condition (3) is not satisfied, then Condition (2) must be satisfied. The converse statement is also true. To see this, suppose Condition (2) is violated. If either (2a) or (2b) is violated, Condition (3) must be satisfied. If neither (2a) nor (2b) is violated, then the violation of Condition (2) requires  $q_r - c_{vr} \geq \zeta_{vr}$ , which implies Condition (3) is satisfied.

The other type of artificial scarcity may occur whenever the integrator leaves a ride unmatched while setting its price below the fare. This is potentially problematic because, by lowering the price to boost its own profit, the integrator undermines the prospect of finding a platform willing to serve that ride. Only by raising the price up to the fare (the upper bound) can we ensure no such platform exists. We name this second type weak artificial scarcity. As we shall see, it is indeed weaker than the first type.

**Definition 2 (Weak Artificial Scarcity).** A given allocation and price vector  $[x, q]$  introduces weak artificial scarcity into the RAP if and only if  $\exists$  a ride  $r$  such that the following condition is violated:

$$\left(1 - \sum_{v \in \mathcal{V}} x_{vr}\right) (q_r - \tilde{q}_r) \geq 0. \quad (4)$$

The definition simply states leaving a ride unmatched requires setting its price at the highest possible level, i.e., the fare, or artificial scarcity cannot be completely ruled out. We next show a solution that eliminates weak artificial scarcity must also eliminate strong artificial scarcity.

**Proposition 1.** If a given allocation and price vector  $[x, q]$  satisfies Condition (4), it must also satisfy Condition (3).

**Proof.** If a given allocation and price vector  $[x, q]$  satisfies Condition (4), then for any ride  $r$ , either of the following two must be satisfied: (i)  $\sum_{v \in \mathcal{V}} x_{vr} = 1$ , or (ii)  $\sum_{v \in \mathcal{V}} x_{vr} = 0$  and  $q_r \geq \tilde{q}_r$ . For (i), Constraint (3) is obviously satisfied. For (ii), first note that  $q_r$  will always be set to  $\tilde{q}_r$  since otherwise the integrator will lose money. This implies  $q_r - c_{vr} - \zeta_{vr} = \tilde{q}_r - c_{vr} - \zeta_{vr}$  for any  $v$ . If  $\tilde{q}_r - c_{vr} - \zeta_{vr} < 0$ , then strong artificial scarcity cannot arise as per Definition 2, which implies Condition (3) is satisfied. Otherwise, we must have  $q_r - c_{vr} - \zeta_{vr} \geq 0$ , which means Constraint (3) is satisfied under (ii) for any  $v$ .  $\square$

Since Proposition 1 implies Condition (4) is capable of eliminating both types of artificial scarcity (see Section 6.1 for a numerical illustration), we shall focus on weak artificial scarcity in this study. Hereafter, unless other specified, the term artificial scarcity always refers to Definition 2.

#### 4. Formulation as a Stackelberg game

In this section, we will interpret the ride allocation problem (RAP) introduced in the previous section as a Stackelberg pricing game and formulate it as a bilevel program. RAP has a natural leader-follower structure. The integrator is the leader who dictates the price  $q$  and the platforms are followers who, through competitions guided by the price, “determine” the allocation  $x$ . In what follows, Section 4.1 presents a bilevel program in which artificial scarcity is eliminated. Section 4.2 reformulates the bilevel program by introducing a virtual platform, while leaving out the constraint that rules out artificial scarcity. We then establish the equivalence between the transformed program and the original bilevel program.

##### 4.1. Original bilevel program

We model the interaction between the integrator and its associated platforms as a Stackelberg game, formulated as a bilevel program. At the upper level, the integrator chooses  $q$  to maximize its profit, i.e.,

Leader’s Problem: Profit maximization for the integrator

$$\max_q L(q) = \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}} (\tilde{q}_r - q_r) x_{vr} \quad (5a)$$

subject to:

$$\sum_{v \in \mathcal{V}} x_{vr} \leq 1, \forall r \in \mathcal{R}, \quad (5b)$$

$$\sum_{r \in \mathcal{R}} x_{vr} \leq 1, \forall v \in \mathcal{V}, \quad (5c)$$

$$\left(1 - \sum_{v \in \mathcal{V}} x_{vr}\right) (q_r - \tilde{q}_r) \geq 0, \forall r \in \mathcal{R}, \quad (5d)$$



$$x_{vr} \in \{0, 1\}, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}, \quad (5e)$$

where the decision variable  $x$  in the objective function (5a), in addition to satisfying Constraints (5b)–(5e), must also be the solution to the follower's problem for each  $p \in \mathcal{P}$ , which reads

Follower's problem: Profit optimization for platform  $p \in \mathcal{P}$

$$\max_{x_p} F(x_p) = \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_p} (q_r - c_{vr}) x_{vr} \quad (6a)$$

subject to:

$$\sum_{v \in \mathcal{V}_p} x_{vr} \leq 1, \forall r \in \mathcal{R}, \quad (6b)$$

$$\sum_{r \in \mathcal{R}} x_{vr} \leq 1, \forall v \in \mathcal{V}_p, \quad (6c)$$

$$q_r \geq (c_{vr} + \zeta_{vr}) x_{vr}, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (6d)$$

$$x_{vr} \in \{0, 1\}, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p. \quad (6e)$$

Constraints (5b)–(5c) and (6b)–(6c) are standard constraints from the bipartite matching problem, and constraints (5e) and (6e) require the matching variables  $x_{vr}$  be binary. Constraint (5d) eliminates weak artificial scarcity, hereafter referred to as the Elimination of Artificial Scarcity, or EAS, constraint. Constraint (6d) ensures the profit of every matched pair is no less than the reserve profit.

In this Stackelberg game, the integrator sets the ride price to coordinate the allocation decisions made by the self-interested platforms. The equilibrium of the game is a state from which no players would have incentives to deviate. We shall refer to Problems (5)–(6) as the original RAP and use  $x^*$  and  $q^*$  to denote, respectively, the allocation decision and pricing vector at the equilibrium.

**Remark 1.** We note that Problems (5)–(6) may not always have a feasible solution. Suppose, for example, the integrator receives only one ride, and two active platforms each operate one idle vehicle. If the reserve price of each platform is identical and nonzero, and the delivery cost of each vehicle is also the same for that ride, then the model may not have a feasible solution for the following reason. No matter which platform receives the ride (thus making a positive profit), the other platform would be unsatisfied (i.e., the solution would not be optimal for them). Obviously, not assigning the ride to any of them is not optimal either. There are at least two ways to rule out this infeasibility: (i) set the reserve price to zero (which will push down the profit to zero, making the platforms indifferent to the receipt of the ride); and (ii) for any ride, ensure no more than one idle vehicle, each from a different platform, has the same value of  $c_{vr} + \zeta_{vr}$  (delivery cost plus reserve profit). In any case, such infeasibility is likely a rare occurrence in practical scenario, as we did not find a single instance among hundreds of random experiments.

#### 4.2. Transformed bilevel program

In this section, we will transform the original RAP to render the hard EAS constraints in the leader's problem obsolete. We begin by giving the following assumption and demonstrate how it can be met by altering RAP.

At first glance, this assumption seems unreasonable since it cannot hold whenever there are more ride requests than vehicles, i.e.,  $|\mathcal{R}| > |\mathcal{V}|$ . To address this issue, we propose to add a virtual platform to the coalition, denoted as  $\bar{p}$ , that owns just enough vehicles to serve all rides, i.e.,  $|\mathcal{R}| = |\mathcal{V}_{\bar{p}}|$ . In order that the virtual platform is eligible to serve any unmatched ride, the cost of each matching pair  $(v, r)$ ,  $v \in \mathcal{V}_{\bar{p}}$  must be no more than the fare, i.e.,  $c_{vr} \leq \bar{q}_r$ ,  $v \in \mathcal{V}_{\bar{p}}$ . Otherwise, the virtual platform could lose money by serving  $r$  with  $v$ , which contradicts the assumption that every platform attempts to maximize profit. On the other hand, the virtual platform must not earn a positive profit from serving any ride, or it might distort the original problem by directly competing with real platforms. This means it will set reserve price as zero ( $\zeta_{\bar{p}} = 0$ ), which leads to  $c_{vr} \geq \bar{q}_r$ ,  $v \in \mathcal{V}_{\bar{p}}$ . The above analysis suggests that the cost of every matching pair  $(v, r)$  must be set to  $c_{vr} = \bar{q}_r$ , for any  $v \in \mathcal{V}_{\bar{p}}$ .

**Assumption 1 (Sufficient Supply).** The integrator is able to serve any ride  $r \in \mathcal{R}$ , i.e.,

$$\sum_{v \in \mathcal{V}} x_{vr} = 1, \forall r \in \mathcal{R}. \quad (7)$$

We are now ready to present a new bilevel program based on the following transformation of RAP.

1. Add into  $\mathcal{P}$  a virtual platform  $\bar{p}$  that owns  $|\mathcal{R}|$  vehicles, and set  $\zeta_{\bar{p}} = 0$  and  $c_{vr} = \bar{q}_r$  for each  $v \in \mathcal{V}_{\bar{p}}$ .
2. Set  $\bar{\mathcal{P}} = \mathcal{P} \cup \{\bar{p}\}$  and  $\bar{\mathcal{V}} = \mathcal{V} \cup \mathcal{V}_{\bar{p}}$ , and replace  $\mathcal{P}$  and  $\mathcal{V}$  in the bilevel program (5)–(6) with  $\bar{\mathcal{P}}$  and  $\bar{\mathcal{V}}$ , respectively.
3. Replace constraint (5b) with the following

$$\sum_{v \in \bar{\mathcal{V}}} x_{vr} = 1, \forall r \in \mathcal{R}. \quad (8)$$

To distinguish between the transformed bilevel program and the original bilevel program, let  $\bar{x} = \{\bar{x}_{vr}\}$  denote the allocation decision after the above transformation process, where  $\forall r \in \mathcal{R}$  and  $\forall v \in \bar{\mathcal{V}}$ . Whereas  $x$  has a dimension of  $|\mathcal{V}| \times |\mathcal{R}|$ , the dimension of  $\bar{x}$  is  $(|\mathcal{V}| + |\mathcal{R}|) \times |\mathcal{R}|$ . Similarly, we introduce  $\bar{c}$  and  $\bar{\zeta}$ , which has the same dimension as  $\bar{x}$ , to replace  $c$  and  $\zeta$  in the original formulation as the cost matrix in the transformed formulation. Also, we introduce  $\bar{x}_p = \{\bar{x}_{vr}, r \in \mathcal{R}, v \in \mathcal{V}_p\}$  and  $\bar{\mathcal{P}} = \{\bar{x}_p, p \in \bar{\mathcal{P}}\}$ .

We write the formulation of the transformed bilevel program using the new notations in the following.

Reformulated leader's problem

$$\max_{\bar{q}} L(\bar{q}) = \sum_{r \in \mathcal{R}} \sum_{v \in \bar{\mathcal{V}}} (\bar{q}_r - \bar{q}_r) \bar{x}_{vr} = \sum_{r \in \mathcal{R}} (\bar{q}_r - \bar{q}_r) \quad (9a)$$

subject to:

$$\sum_{v \in \bar{\mathcal{V}}} \bar{x}_{vr} = 1, \forall r \in \mathcal{R}, \quad (9b)$$

$$\sum_{r \in \mathcal{R}} \bar{x}_{vr} \leq 1, \forall v \in \bar{\mathcal{V}}, \quad (9c)$$

$$\left(1 - \sum_{v \in \bar{\mathcal{V}}} \bar{x}_{vr}\right) (\bar{q}_r - \bar{q}_r) \geq 0, \forall r \in \mathcal{R}, \quad (9d)$$

$$\bar{x}_{vr} \in \{0, 1\}, \forall r \in \mathcal{R}, \forall v \in \bar{\mathcal{V}}, \quad (9e)$$

where  $\bar{x}$  is the solution to the follower's problem for each platform  $p \in \bar{\mathcal{P}}$ , i.e.,

Reformulated follower's problem for platform  $p \in \bar{\mathcal{P}}$

$$\max_{\bar{x}_p} F(\bar{x}_p) = \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_p} (\bar{q}_r - \bar{c}_{vr}) \bar{x}_{vr} \quad (10a)$$

subject to:

$$\sum_{v \in \mathcal{V}_p} \bar{x}_{vr} \leq 1, \forall r \in \mathcal{R}, \quad (10b)$$

$$\sum_{r \in \mathcal{R}} \bar{x}_{vr} \leq 1, \forall v \in \mathcal{V}_p, \quad (10c)$$

$$\bar{q}_r \geq (\bar{c}_{vr} + \bar{\zeta}_{vr}) \bar{x}_{vr}, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (10d)$$

$$\bar{x}_{vr} \in \{0, 1\}, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p. \quad (10e)$$

In the transformed problem, the integrator's profit is no longer affected by the allocation decision (see Eq. (9a)). This is because by construction all requests must be served and any requests served by the virtual platform would yield zero profit. Thus, the upper level objective in the transformed problem is reduced from a bilinear function to a linear one. Indeed, any ride request matched with the virtual platform  $\bar{p}$  in the transformed formulation would in fact be rejected by the integrator in the original formulation.

The following result establishes the equivalence between the original and the transformed bilevel programs.

**Proposition 2.** Let  $[\bar{x}^*, \bar{q}^*]$  be an optimal solution to the bilevel program (9)–(10), and construct a feasible solution to the bilevel program (5)–(6), denoted as  $[x^*, q^*]$ , such that

$$x_{vr}^* = \bar{x}_{vr}^*, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}, \quad (11a)$$

$$q_r^* = \bar{q}_r^*, \forall r \in \mathcal{R}. \quad (11b)$$

$[x^*, q^*]$  is an optimal solution to the bilevel program (5)–(6).

**Proof.** See Appendix B.  $\square$

The next result asserts we may remove the EAS constraint from the transformed bilevel program without affecting the optimal solution.

**Proposition 3.** The EAS constraints (Eq. (9d)) is redundant in the transformed bilevel program (9)–(10).

**Proof.** See Appendix C.  $\square$

With Proposition 3, we show artificial scarcity can be eliminated through the transformation introduced in this section, without using the clumsy EAS constraint. This reduces the original RAP to a bilevel pricing problem that has a linear upper-level objective and a bilinear lower-level objective. Hereafter, RAP refers to the transformed bi-level pricing problem (9)–(10), rather than the original version. For the purpose of comparison, we also consider a ride allocation problem that allows artificial scarcity, which is formulated as Problem (5)–(6) without the EAS constraint. This problem will be referred to as RAP-AS in the following.



**Table 1**  
Cost required to deliver ride  $r$  by vehicle  $v$  and ride fare  $\tilde{q}$  paid to the integrator.

|        | $\tilde{q}$ | $c_{vr}$               |                        |                        |
|--------|-------------|------------------------|------------------------|------------------------|
|        |             | Vehicle 1 (platform 1) | Vehicle 2 (platform 2) | Vehicle 3 (platform 3) |
| Ride 1 | 6           | 3                      | 1                      | 4                      |
| Ride 2 | 11          | 4                      | 1                      | 4                      |
| Ride 3 | 7           | 1                      | 2                      | 3                      |

**Table 2**  
Matched pairs from the optimal solution to RAP-AS (left panel) and RAP (right panel) for the illustrative example.

| (a) With artificial scarcity |        |        |        | (b) Without artificial scarcity |        |        |        |
|------------------------------|--------|--------|--------|---------------------------------|--------|--------|--------|
| $x^*$                        | Ride 1 | Ride 2 | Ride 3 | $x^*$                           | Ride 1 | Ride 2 | Ride 3 |
| Vehicle 1                    | 0      | 0      | 1      | Vehicle 1                       | 0      | 0      | 1      |
| Vehicle 2                    | 0      | 1      | 0      | Vehicle 2                       | 1      | 0      | 0      |
| Vehicle 3                    | 0      | 0      | 0      | Vehicle 3                       | 0      | 1      | 0      |

**Table 3**  
Price, revenue and profit from the optimal solution to RAP-AS (the second-last row) and RAP (the last row) for the illustration example.

|                             | Price $q^*$ |        |        | Integrator        |                  |                |
|-----------------------------|-------------|--------|--------|-------------------|------------------|----------------|
|                             | Ride 1      | Ride 2 | Ride 3 | Fare              | Payment          | Profit         |
| With artificial scarcity    | 0           | 1      | 1      | $11 + 7 = 18$     | $1 + 1 = 2$      | $18 - 2 = 16$  |
| Without artificial scarcity | 4           | 4      | 2      | $6 + 11 + 7 = 24$ | $4 + 4 + 2 = 10$ | $24 - 10 = 14$ |

### 4.3. Illustrative example

We next provide an illustrative example of the integrator's ride allocation problem. Consider an integrator who receives three rides in a decision episode and has three active platforms that each operates one idle vehicle. The fare  $\tilde{q}$  and the cost matrix  $c$  are given in Table 1. For simplicity, we assume all platforms set a zero reserve price, i.e.  $\zeta_p = 0, \forall p$ .

To demonstrate the impact of artificial scarcity, Tables 2 and 3 report the optimal allocation, price, and the integrator's profit produced by solving RAP-AS and RAP for the example. For RAP-AS, only the pairs (1, 3) and (2, 2) are matched. The allocation leaves Ride 1 and Vehicle 3 unmatched, leading to artificial scarcity – note the cost for Vehicle 3 to serve Ride 1 is  $c_{31} = 4$ , lower than the fare  $\tilde{q}_1 = 6$ . Enforcing the EAS constraint, as expected, produces three matches: (1, 3), (2, 1) and (3, 2), see Table 2(b).

Table 3 explains why Ride 1 is unmatched in RAP-AS. When Ride 1 is matched, the price set by the integrator is  $q = (4, 4, 2)$  (the last row in Table 3). By removing that ride from the system (hence creating artificial scarcity), the integrator can depress the price to  $q = (0, 1, 1)$  (the second last row in Table 3). This generates a revenue of 18 and a profit of 16. In comparison, RAP yields a higher revenue (24) but a lower profit (14).

## 5. Analysis of RAP

In Section 5.1, we reformulate RAP as a single-level mixed integer linear program (MILP) by exploiting duality. Section 5.2 shows this MILP can be further reduced to a linear program (LP) — hence is polynomially solvable — when all platforms set their reserve price to zero. It also explores the relationship between this particular version of the RAP and the core of the cooperative assignment game (Shapley and Shubik, 1971).

### 5.1. Reformulation as MILP

We first note that Constraint (10d) is equivalent to

$$\begin{cases} \bar{x}_{vr} \in \{0, 1\} & \text{if } \tilde{q}_r \geq \bar{c}_{vr} + \bar{\xi}_{vr}, \\ \bar{x}_{vr} = 0 & \text{if } \tilde{q}_r < \bar{c}_{vr} + \bar{\xi}_{vr}, \end{cases} \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (12)$$

which can be further simplified as

$$\bar{x}_{vr} = 0, \text{ if } \tilde{q}_r < \bar{c}_{vr} + \bar{\xi}_{vr}, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p. \quad (13)$$

By replacing Constraint (10d) with Constraint (13), Problem (10) is turned into a variant of the bipartite matching problem in which certain pairs are precluded according to Constraint (13). Since Problem (10) now has constraints with a coefficient matrix satisfying total unimodularity (Tamir, 1976), it shares the same optimal objective function value with its linear relaxation. This property enables us to invoke the strong duality theorem when integrating the follower's problem with the leader's problem.

Let  $\kappa_p = \{\kappa_{rp}, r \in \mathcal{R}\}$ ,  $\varphi_p = \{\varphi_v, v \in \mathcal{V}_p\}$  and  $\delta_p = \{\delta_{vr}, r \in \mathcal{R}, v \in \mathcal{V}_p\}$  be dual variables corresponding, respectively, to the primal constraints (10b), (10c) and (13); let  $\kappa = \{\kappa_p, p \in \bar{\mathcal{P}}\}$ ,  $\varphi = \{\varphi_p, p \in \bar{\mathcal{P}}\}$  and  $\delta = \{\delta_p, p \in \bar{\mathcal{P}}\}$ . For each platform  $p \in \bar{\mathcal{P}}$ , let first write down the dual of Problem (10) as

$$\min_{\kappa_p, \varphi_p, \delta_p} \sum_{r \in \mathcal{R}} \kappa_{rp} + \sum_{v \in \mathcal{V}_p} \varphi_v \quad (14a)$$

subject to:

$$\kappa_{rp} + \varphi_v + \delta_{vr} \geq \bar{q}_r - \bar{c}_{vr}, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (14b)$$

$$\kappa_{rp}, \varphi_v \geq 0, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (14c)$$

$$\begin{cases} \delta_{vr} = 0 & \text{if } \bar{q}_r \geq \bar{c}_{vr} + \bar{\xi}_{vr}, \\ -\infty \leq \delta_{vr} \leq \infty & \text{if } \bar{q}_r < \bar{c}_{vr} + \bar{\xi}_{vr}, \end{cases} \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p. \quad (14d)$$

Constraint (14d) can be linearized by introducing additional integer variables as follows:

$$-M(1 - z_{vr}) \leq \delta_{vr} \leq M(1 - z_{vr}), \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (15a)$$

$$\frac{\bar{q}_r}{\bar{c}_{vr} + \bar{\xi}_{vr}} \geq 1 - M(1 - y_{vr}), \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (15b)$$

$$\frac{\bar{q}_r}{\bar{c}_{vr} + \bar{\xi}_{vr}} < 1 + M(1 - y_{vr}), \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (15c)$$

$$y_{vr} + z_{vr} = 1, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (15d)$$

$$y_{vr}, z_{vr} \in \{0, 1\}, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \quad (15e)$$

where  $M > 0$  is a very large real number. To see the equivalence with Constraint (14d), verify that (15) only has two outcomes:  $y_{vr} = 0$  and  $z_{vr} = 1$  implies  $\bar{q}_r \geq \bar{c}_{vr} + \bar{\xi}_{vr}$  and  $\delta_{vr} = 0$ ; and  $y_{vr} = 1$  and  $z_{vr} = 0$  implies  $\bar{q}_r < \bar{c}_{vr} + \bar{\xi}_{vr}$  and  $-\infty \leq \delta_{vr} \leq \infty$ .

The strong duality theorem state that if  $[\kappa_p, \varphi_p, \delta_p]$  and  $\bar{x}_p$  are the optimal solution to Problem (14) and (10) respectively, then we have

$$\sum_{r \in \mathcal{R}} \kappa_{rp} + \sum_{v \in \mathcal{V}_p} \varphi_v = \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_p} (\bar{q}_r - \bar{c}_{vr}) \bar{x}_{vr}. \quad (16)$$

We can then convert RAP into a single-level optimization problem by including in the leader's problem the dual feasibility conditions and the strong duality condition (Kleinert and Schmidt, 2021).

$$\max_{\bar{x}, y, z, \bar{q}, \kappa, \varphi, \delta} \sum_{r \in \mathcal{R}} (\bar{q}_r - \bar{q}_r) \quad (17a)$$

subject to:

$$(9b), (9c), (9e), \quad (17b)$$

$$(10d), (14b), (14c), (15), (16), \forall p \in \bar{\mathcal{P}}. \quad (17c)$$

Problem (17) is a mixed integer nonlinear program, since the strong duality constraint (16) contains a bilinear term  $\bar{q}_r \bar{x}_{vr}$ . We next show this constraint can be reduced to a linear one.

**Proposition 4.** A solution vector  $[\kappa, \varphi, x]$  satisfy Constraint (16) for all  $p \in \bar{\mathcal{P}}$  if and only if it also satisfies the following single constraint

$$\sum_{p \in \bar{\mathcal{P}}} \left( \sum_{r \in \mathcal{R}} \kappa_{rp} + \sum_{v \in \mathcal{V}_p} \varphi_v \right) = \sum_{r \in \mathcal{R}} \bar{q}_r - \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}} \bar{c}_{vr} \bar{x}_{vr}. \quad (18)$$

**Proof.** See Appendix D.  $\square$

Using Proposition 4, we can reformulate RAP as a mixed-integer linear program (MILP).

$$\max_{\bar{x}, y, z, \bar{q}, \kappa, \varphi, \delta} \sum_{r \in \mathcal{R}} (\bar{q}_r - \bar{q}_r) \quad (19a)$$

subject to:

$$(9b), (9c), (9e), (18), \quad (19b)$$

$$(10d), (14b), (14c), (15), \forall p \in \bar{\mathcal{P}}. \quad (19c)$$

### 5.2. Special case: zero reserve price

We proceed to consider a special version of Problem (19) for which stronger results are available. In this version, all platforms set their reserve price to zero, i.e.,  $\zeta_p = 0$  for all  $p$ . Note that this restriction means Constraint (10d) is no longer needed, as a ride can be accepted as long as it provide a zero profit (rather than a positive reserve profit). Accordingly, the dual variables  $\delta$  are removed from Constraint (14d). With this change, Problem (19) now reads

$$\max_{\bar{x}, \bar{q}, \kappa, \varphi} \sum_{r \in \mathcal{R}} (\bar{q}_r - \bar{q}_r) \quad (20a)$$

subject to:

$$(9b), (9c), (9e), (18) \quad (20b)$$

$$(14c), \forall p \in \bar{\mathcal{P}}, \quad (20c)$$

$$\kappa_{rp} + \varphi_v \geq \bar{q}_r - \bar{c}_{vr}, \forall r \in \mathcal{R}, \forall v \in \mathcal{V}_p, \forall p \in \bar{\mathcal{P}}. \quad (20d)$$

The following theorem establishes another main result of this paper, that Problem (20) – or equivalently, the bilevel Problem (9)–(10) with  $\bar{\zeta}_{vr} = 0, \forall v, r$  — is polynomially solvable.

**Theorem 1.** Construct a linear relaxation of Problem (20) as follows:

$$\max_{\bar{x}, \bar{q}, \kappa, \varphi} \sum_{r \in \mathcal{R}} (\bar{q}_r - \bar{q}_r) \quad (21a)$$

subject to:

$$(9b), (9c), (18), (20c) (20d) \quad (21b)$$

$$0 \leq x_{vr} \leq 1, \forall r \in \mathcal{R}, \forall v \in \bar{\mathcal{V}}. \quad (21c)$$

The optimal objective function values of Problems (20) and (21) are equal.

**Proof.** See Appendix E.  $\square$

The proof of Theorem 1 reveals that the single-level problem (20) can be decomposed into two subproblems. The first is the bipartite matching problem (34), which can be solved to obtain the optimal allocation of problem (20), denoted as  $x^*$ , independent of ride price ( $\bar{q}$ ) and multipliers ( $\kappa$  and  $\varphi$ ). This is because Constraint (18) in Problem (20) is incorporated into the objective function of (34), and Constraints (20c) and (20d) do not interact with the allocation.

With the optimal allocation  $x^*$ , the second subproblem determines the optimal price and multipliers by solving another reduced form of Problem (20), i.e.,

$$\max_{\bar{q}, \kappa, \varphi} \sum_{r \in \mathcal{R}} \eta_r \equiv \sum_{r \in \mathcal{R}} (\bar{q}_r - \bar{q}_r) \quad (22a)$$

subject to:

$$\sum_{r \in \mathcal{R}} \eta_r + \sum_{p \in \bar{\mathcal{P}}} \sum_{r \in \mathcal{R}} \kappa_{rp} + \sum_{v \in \bar{\mathcal{V}}} \varphi_v = \sum_{r \in \mathcal{R}} \sum_{v \in \bar{\mathcal{V}}} (\bar{q}_r - \bar{c}_{vr}) x_{vr}^*, \quad (22b)$$

$$(20c), (20d), \quad (22c)$$

where  $\eta_r$  is a new variable introduced to represent the revenue generated for the integrator by serving ride  $r$ , and Constraint (22b) is obtained by embedding the optimal allocation in Constraint (18) and noticing

$$\sum_{r \in \mathcal{R}} \eta_r + \sum_{p \in \bar{\mathcal{P}}} \sum_{r \in \mathcal{R}} \kappa_{rp} + \sum_{v \in \bar{\mathcal{V}}} \varphi_v = \sum_{r \in \mathcal{R}} \bar{q}_r - \sum_{r \in \mathcal{R}} \sum_{v \in \bar{\mathcal{V}}} \bar{c}_{vr} x_{vr}^*.$$

We may interpret the right hand side (RHS) of Constraint (20d) as the extra profit generated by serving ride  $r$  with vehicle  $v$  owned by platform  $p$ , whereas  $\kappa_{rp}$  and  $\varphi_v$  on the left hand side (LHS) are the share of this profit assigned to, respectively, the integrator and the platform  $p$ . The complementarity-slackness condition guarantees that, for any realized match, the two sides of the constraint must be equal. Similarly, the RHS of Constraint (22b) represents the total profit generated by the system less the income of the platforms earned from reserve profit. The first two terms on the LHS of that constraint can be viewed as the share earned by the integrator and the third is that by the platforms.

Let us define

$$\bar{\eta}_r = \eta_r + \sum_{p \in \bar{\mathcal{P}}} \kappa_{rp} \quad (23)$$

as the total profit the integrator receives from serving ride  $r$ , where  $\eta_r$  derives from the price spread and  $\sum_{p \in \bar{\mathcal{P}}} \kappa_{rp}$  is the integrator's share in the platform's extra profit. We next show that at the optimum RAP would not give any extra profit to the integrator in addition to the price spread. That is,  $\kappa = 0$  and  $\bar{\eta}_r = \eta_r$ .

**Table 4**  
Vehicle ownership and profit in the seven-ride example.

|                   |                             | Real platform |      |      | Virtual platform | integrator |
|-------------------|-----------------------------|---------------|------|------|------------------|------------|
|                   |                             | 1             | 2    | 3    | 4                |            |
| Vehicle ownership |                             | 1–2           | 3–4  | 5–6  | 7–13             | –          |
| Profit            | With artificial scarcity    | 0             | 0    | 0.49 | –                | 33.79      |
|                   | Without artificial scarcity | 0.11          | 0.85 | 1.37 | 0                | 33.16      |

**Proposition 5.** If  $[\eta^*, \kappa^*, \varphi^*]$  is an optimal solution of Problem (22), then  $\kappa^* = 0$ .

**Proof.** See Appendix F  $\square$

Problem (22) may be viewed as the core of an collaborative assignment game (Shapley and Shubik, 1971). A classical assignment game is a real estate market that connects a set of homeowners, who set the home price, to a set of potential buyers, who assign different valuations for each home. In RAP, the setting is reversed since the buyers (i.e., the riders) sets the price ( $\tilde{p}_r$ ) and the sellers (i.e., the vehicles) assign a cost to each ride ( $c_{vr}$ ). The profit (called “worth” in Shapley and Shubik, 1971) of each matched pair is  $\max(0, (\tilde{p}_r - \bar{q}_r) + (\bar{q}_r - c_{vr}))$  with  $\tilde{p}_r - \bar{q}_r$  and  $\bar{q}_r - c_{vr}$  being the share of the profit enjoyed by the rider and the vehicle respectively at the current price  $\bar{q}_r$ . A socially optimal allocation of the assignment game maximizes the total profit, or social welfare, of all matched pairs, i.e.

$$S = \sum_{v \in \mathcal{V}} \sum_{r \in \mathcal{R}} (\tilde{p}_r - c_{vr}) x_{vr}. \quad (24)$$

With the virtual platform (per Assumption 1), all rides that no real vehicles want to serve will be served. Thus, maximizing  $S$  would equal minimizing the total cost  $\sum_{v \in \mathcal{V}} \sum_{r \in \mathcal{R}} c_{vr} x_{vr}$ , which is the objective function of Problem (34). Because Problem (22) maximizes the profit for the buyers (collectively represented by the integrator), it is a buyer-optimal core that reserves as much profit for the integrator as possible.

The above analysis also indicates that, when both artificial scarcity and reserve price are eliminated, RAP, a bilevel pricing problem in which the integrator and the platforms each maximize their own profits, is equal to a cooperative assignment game that maximizes the system’s social welfare. On the flip side, introducing either artificial scarcity or reserve price likely causes the social welfare to deviate from the social optimal value, and makes the problem intractable. In the next section, we will examine social welfare varies with both mechanisms.

## 6. Numerical experiments

We conduct two sets of experiments. In the first (Section 6.1), we construct a small example to investigate the relationship between strong and weak artificial scarcity and how artificial scarcity and reserve price affect allocation results. Section 6.2 reports the results of a case study designed to test the sensitivity of the allocation problem to such factors as market thickness, price level (both ride prices and reserve price) and supply–demand imbalance.

To benchmark RAP, we solve RAP-AS (i.e., the ride allocation problem with artificial scarcity, defined at the end of Section 4.2). Both problems are solved using the MILP solver `intlinprog` in Gurobi. The algorithms are coded in Matlab R2018a. All numerical results are obtained on a laptop with Inter(R) E5-1620 v4 CPU at 3.50 GHz and 32G RAM.

### 6.1. Small example

We consider a ride-hail application with seven rides and three platforms that each own two vehicles, referred to as the seven-ride example. We first set the reserve price as 0. Since the total number of the vehicles (six) is less than the number of rides, at least one ride will not be served. To eliminate artificial scarcity, we add a virtual platform  $\bar{p}$  owning seven vehicles and set the delivery cost  $\bar{c}_{vr} = \bar{q}_r, \forall v \in \mathcal{V}_{\bar{p}}$ . Table 4 maps vehicles, including both real and virtual ones, to the platforms. The fare  $\bar{q}$  of each ride and the delivery cost  $c$  of every pair  $(v, r)$  are randomly generated and reported in Figs. 2(a) and 2(b).

Fig. 2(c–d) and (e–f) visualize, respectively, the optimal solution to RAP-AS and RAP of the seven-ride example. For RAP-AS, the allocation leaves Vehicle 1, Ride 2 and Ride 7 unmatched (See Fig. 2(d)). Combining Figs. 2(a)–2(c), we can see  $q_2 = 2.55 \times 0.19 = 0.4845 < \bar{q}_2 = 2.55 < c_{12} = 5.62$  and  $q_7 = 2.55 \times 0.19 = 0.4845 < \bar{q}_7 = 1.72 < c_{17} = 2.42$ , which indicate no strong artificial scarcity exists. However, the integrator has created weak artificial scarcity, because it sets the price on Ride 2 at only 19% of the fare. Thus, in RAP-AS, the optimal allocation is weakly artificial scarce, but not strongly artificial scarce. Furthermore, the integrator grabs more than a lion’s share of all profit here: 33.79 vs. 0.49 earned by platform 1 (the other two platforms earned nothing!) when weak artificial scarcity exists (See Table 4). This is expected given the integrator’s dominant position as the resource allocator.

For RAP, the optimal allocations are slightly altered. Specifically, Vehicle 4 is switched from Ride 4 to Ride 2 and Vehicle 1 (which is previously left idle) is matched with Ride 4. Since now the only ride (Ride 7) not matched with a real vehicle is matched with the virtual platform (hence being priced at the level of its fare), weak artificial scarcity is eliminated. The integrator is forced to raise the price on all rides but Ride 5. Accordingly, the share of the platforms’ profit in the total profit rises five-fold, at the expense

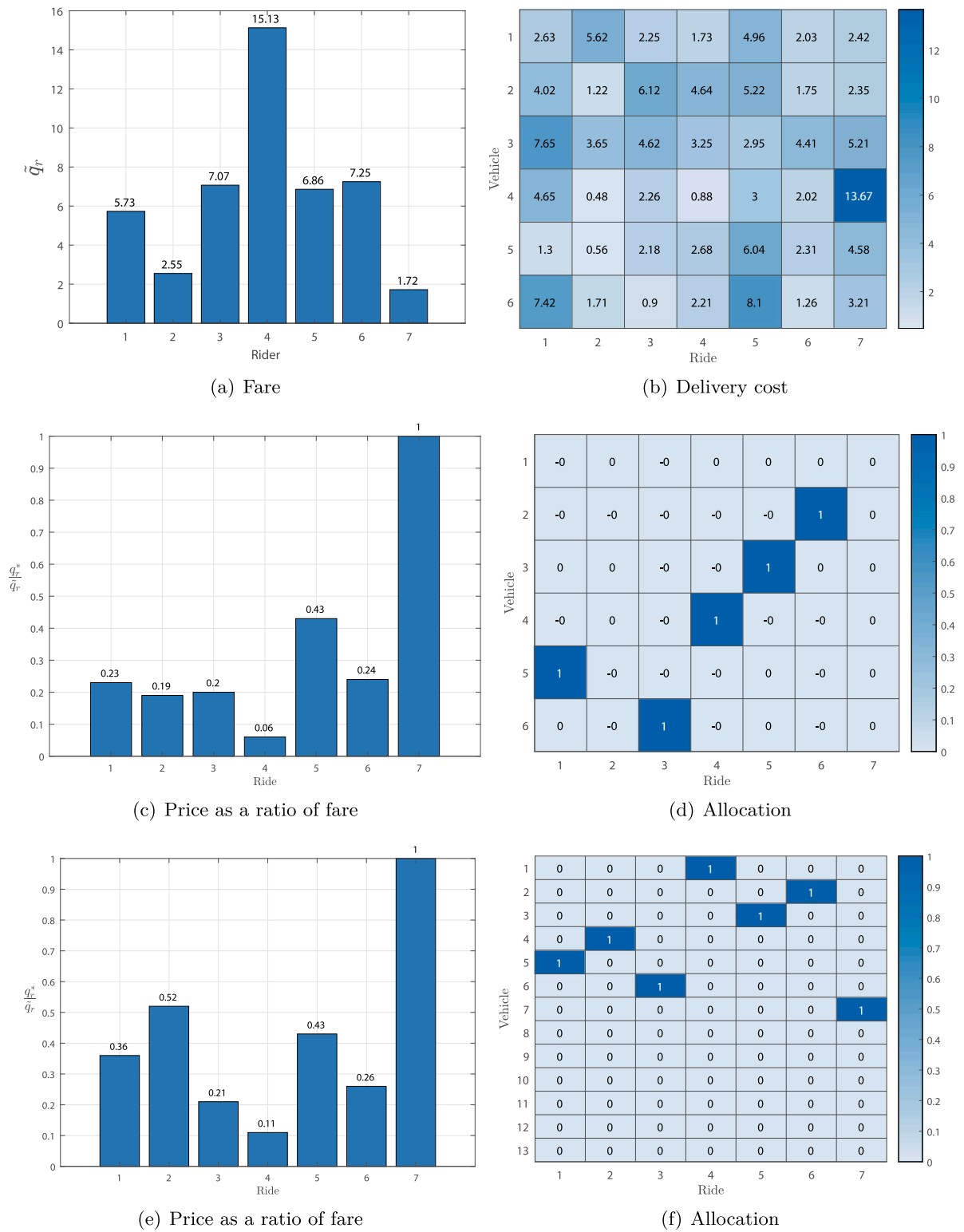


Fig. 2. Visualization of the inputs and optimal solutions to the seven-ride problem. (a–b) inputs; (c–d) optimal solution to RAP-AS; (e–f) optimal solution to RAP.

**Table 5**  
Profit in the seven-ride example with varying reserve profits.

|            | Reserve profits | Profit (Number of matched pairs) |          |          | Integrator | Social welfare |
|------------|-----------------|----------------------------------|----------|----------|------------|----------------|
|            |                 | 1                                | 2        | 3        |            |                |
| With AS    | (3, 3, 3)       | 3 (1)                            | 6 (2)    | 6.49 (2) | 18.79      | 34.28          |
|            | (5, 5, 5)       | 5 (1)                            | 5 (1)    | 5.49 (1) | 10.44      | 25.93          |
|            | (7, 7, 7)       | 0 (0)                            | 7 (1)    | 0 (0)    | 7.25       | 14.25          |
|            | (3, 5, 7)       | 6 (2)                            | 0 (0)    | 0 (0)    | 12.91      | 18.91          |
|            | (5, 7, 3)       | 0 (0)                            | 0 (0)    | 6.47 (2) | 12.15      | 18.62          |
|            | (7, 3, 5)       | 0 (0)                            | 6.64 (2) | 5 (1)    | 12.70      | 24.34          |
| Without AS | (3, 3, 3)       | 3 (1)                            | 6 (2)    | 6.49 (2) | 18.79      | 34.28          |
|            | (5, 5, 5)       | 5 (1)                            | 5 (1)    | 5.49 (1) | 10.44      | 25.93          |
|            | (7, 7, 7)       | 0 (0)                            | 7 (1)    | 0 (0)    | 7.25       | 14.25          |
|            | (3, 5, 7)       | 7.15 (2)                         | 5 (1)    | 0 (0)    | 12.44      | 24.59          |
|            | (5, 7, 3)       | 5 (1)                            | 0 (0)    | 9.92 (2) | 9.59       | 24.51          |
|            | (7, 3, 5)       | 0 (0)                            | 8.23 (2) | 6 (1)    | 10.11      | 24.34          |

of the integrator (See Table 4). Clearly, eliminating artificial scarcity not only helps riders but also boosts the platforms' bottom-line, because it weakens the integrator's monopoly power. However, even in this case, the integrator still takes an overwhelming majority of the profit (nearly 95%). Again, this result is not surprising, since we have shown in the previous section that, when reserve price is zero, RAP is equivalent to the buyer-optimal core of a cooperative assignment game. That is, by design the profit of the buyer (i.e., the integrator) is always prioritized.

We next allow each platform to charge a reserve profit<sup>7</sup> and observe how different charging schemes affect the distribution of profits between the platforms and the integrator. We consider six schemes classified into two types: homogeneous schemes with the same reserve profit applied to all platforms (3, 5, or 7), and heterogeneous schemes in which each platform charges a different reserve profit (again 3, 5, or 7), see Table 5.

When the reserve profit is the same across the platforms, the higher the reserve profit, the less the profit for the integrator, and the lower the social welfare (total profit). At the lowest charging level ( $\zeta = 3$ ), the matching result remains intact. In this case, the integrator simply covers the reserve profit from its own profit, which costs 15 whether artificial scarcity is allowed or not. At the highest level ( $\zeta = 7$ ), the integrator is forced to deny most rides, and gives the only viable ride to platform 2. As a result, both social welfare and the integrator's profit plummet, and nearly half of the total profit now goes to platform 2. Interestingly, in all three cases with the uniform reserve profit, eliminating artificial scarcity makes no difference at all.

The outcome is quite different when each platform charges a different reserve profit. First, the most expensive platform—the one that charges the highest fee—never receives a ride, and the cheapest platform always get most rides, sometimes all rides. Second, eliminating artificial scarcity increases the number of matching pairs from 2 to 3 when the reserve profits for platforms 1, 2, and 3 are set to (3, 5, 7) and (5, 7, 3).

For a platform, the reserve profit is a double-edged sword. When properly set, it can greatly improve profitability. However, an overly aggressive strategy could backfire as the integrator can always take its business elsewhere. Excessively high reserve profits are also detrimental to the system as a whole, because they can undermine the matching productivity. For the integrator, the platforms' demand for reserve profits could only be met by shedding its own profit. However, this is not necessarily a zero-sum game, because in many cases the integrator may still be better off by keeping a platform that demands a sizable reserve profit in the system than losing it to a competitor. Thus, reserve profits may be used as an effective tool to check the monopoly power of the integrator.

## 6.2. Case study

Our case study is set in a square region with side  $L$ . The default value of  $L$  is 1.5 km. To create an instance of the allocation problem, we let the origin of each ride and the location of each idle vehicle uniformly distributed in the region. Let  $s_{vr}$  denote the travel distance from vehicle  $v$ 's location to ride  $r$ 's origin, which is calculated by the Euclidean distance between the two points multiplied by a detour scalar  $\gamma = 1.2$ . The travel time of ride  $r$ , denoted as  $t_r$ , is drawn from a Log-Normal distribution with parameters  $(\mu, \sigma^2) = (2.5, 0.4)$ . This corresponds to a mean travel time of 13.18 min and a standard deviation of 5.47 min, as reported in Chen et al. (2017) using ride-hail data collected in Hangzhou, China. We set the average travel speed  $u = 25$  km/h, which is close to the prevailing speed observed in large Chinese cities in the past decade.<sup>8</sup> Accordingly, the travel distance of ride  $r$  is given by  $s_r = ut_r$ . If vehicle  $v$  is matched with ride  $r$ , it will take the vehicle a total time of  $t_{vr} = \frac{s_{vr}}{u} + t_r$  to complete the trip. Following the standard practice in China, we compute the trip fare based on both travel distance and travel time. Let  $\alpha$  be the price per unit distance and  $\beta$  the price per unit travel time. We set  $\alpha = ¥1.6/\text{km}$  and  $\beta = ¥0.5/\text{min}$  based on DidiChuxing's fare schedule.<sup>9</sup> Thus, the fare  $\bar{q}_r = \alpha s_r + \beta t_r$ . To estimate the trip cost, we set the wage rate  $g = ¥22/\text{h}$  (the current minimum wage in Chengdu, China<sup>10</sup>) and

<sup>7</sup> Note that, unlike the reserve price, the reserve profit is independent of the transportation cost.

<sup>8</sup> See [https://www.cac.gov.cn/2021-02/03/c\\_1613923423079314.htm](https://www.cac.gov.cn/2021-02/03/c_1613923423079314.htm).

<sup>9</sup> See <http://it.people.com.cn/n1/2019/0710/c1009-31224190.html>.

<sup>10</sup> [http://www.chenghua.gov.cn/chqrmzfw/c144318hgfdqdd/2022-06/09/content\\_0a11bdafb8234f6daaacee30ad33da9c.shtml](http://www.chenghua.gov.cn/chqrmzfw/c144318hgfdqdd/2022-06/09/content_0a11bdafb8234f6daaacee30ad33da9c.shtml).



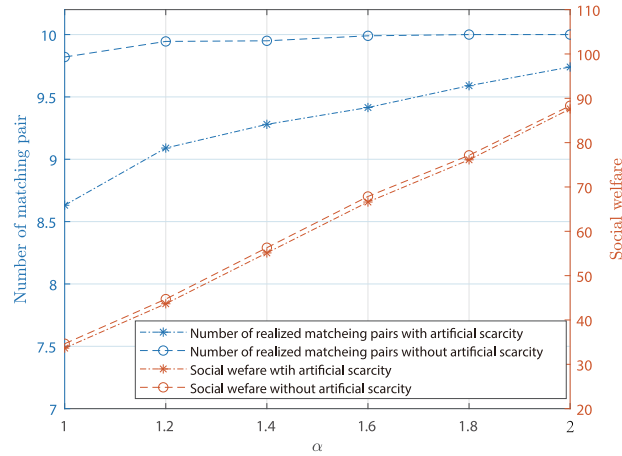


Fig. 3. Effect of  $\alpha$  (fare per unit distance) on the number of matched pairs and social welfare. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the fuel cost  $h = ¥0.8/\text{km}$  (Estimated based on fuel efficiency of a typical passenger sedan (10 km/L) and the gasoline price in China (about ¥8/L)). Thus, the delivery cost of each matching pair is estimated as  $c_{vr} = gt_{vr} + h(s_r + s_{vr})$  (It is possible to make  $g$  and  $h$  platform-specific. That is, these parameters may be set according to the contract negotiated between the integrator and each platform. However, we need not consider this complication here.). Unless otherwise noted, all input parameters will take the default value specified above.

We perform four sets of simulation experiments. The first and the second aim to examine the effects of trip fare and market density, respectively. In both experiments, we set  $|\mathcal{P}| = 2$ ,  $|\mathcal{R}| = 10$  and give each platform five vehicles. Thus, the supply and demand are perfectly balanced in the market and, given the default  $L$  value (1.5 km), there are on average 4.4 rides and 4.4 idle vehicles per square km. The third experiment deals with the impact of supply–demand imbalance, and so the number of rides and vehicles will vary accordingly. In the above three experiments, the reserve price of both platforms are set to zero. The impact of reserve price is addressed in the last experiment.

Throughout this section, all reported results are the average of a sample of 500. That is, we solve 500 instances for each scenario – characterized by a combination of input parameters and artificial scarcity setting – and take the average of the 500 outputs to represent that scenario. Each instance is determined by a set of randomly selected ride and vehicle locations, as well as a set of random ride distances drawn from the Log-Normal distribution.

#### 6.2.1. Trip fare

We use  $\alpha$ , the fare per unit distance, to assess the sensitivity of the allocation results to trip fare. Other parameters, such as  $\beta$  (fare per unit time),  $g$  (cost per unit time) and  $h$  (cost per unit distance), would contribute to the trade-off between price and cost in a similar manner. Fig. 3 indicates raising fare generally improves matching productivity, especially when artificial scarcity exists. This is expected, because a higher fare expands the supply pool. The effect, however, is relatively small. When  $\alpha$  doubles from ¥1 per km to ¥2, the average number of matched pairs increases from 8.63 to 9.72 (a 13% increase), with artificial scarcity. When artificial scarcity is forbidden, the fare has a much smaller effect on matching productivity within the tested range (less than 3%). Since the fare rate affects the revenue but not the cost, it has a much larger effect on social welfare (defined in (24)): doubling the rate more than doubles social welfare. However, the impact of artificial scarcity appears to be negligible in this case: the improvement achieved by eliminating artificial scarcity never exceeds three percent and gradually decreases with  $\alpha$  (see the two red lines in Fig. 3).

Fig. 4 compares the profitability of each stakeholder against the average fare and the average cost. As expected, the integrator is much more profitable—with a per capita profit at least an order of magnitude larger than those of the two platforms—regardless of the status of artificial scarcity. When artificial scarcity is eliminated, the platforms' profitability does rise, but is still much lower than the integrator's. Moreover, while the rising fare benefits the integrator in both cases, it does not benefit the platforms at all. Indeed, Fig. 4(b) shows the per capital profit of the two platforms stay flat as  $\alpha$  increases.

#### 6.2.2. Market density

When the number of rides and vehicles is fixed, the market density depends on the area of the region ( $L^2$ ). In this section, we vary the density by changing  $L$ , while holding the numbers of rides and vehicles constant.

As shown in Fig. 5, reducing the market density (by increasing  $L$ ) affects the number of matched pairs and total social welfare negatively. This is because a more sparse market is associated with a longer average pickup distance and thus a higher average delivery cost. However, the effect is rather small. When  $L$  increases by a factor of five (a drop in density by more than 95%), there is a mere 12% reduction in the matching productivity even with artificial scarcity. Without it, the effect of  $L$  on productivity

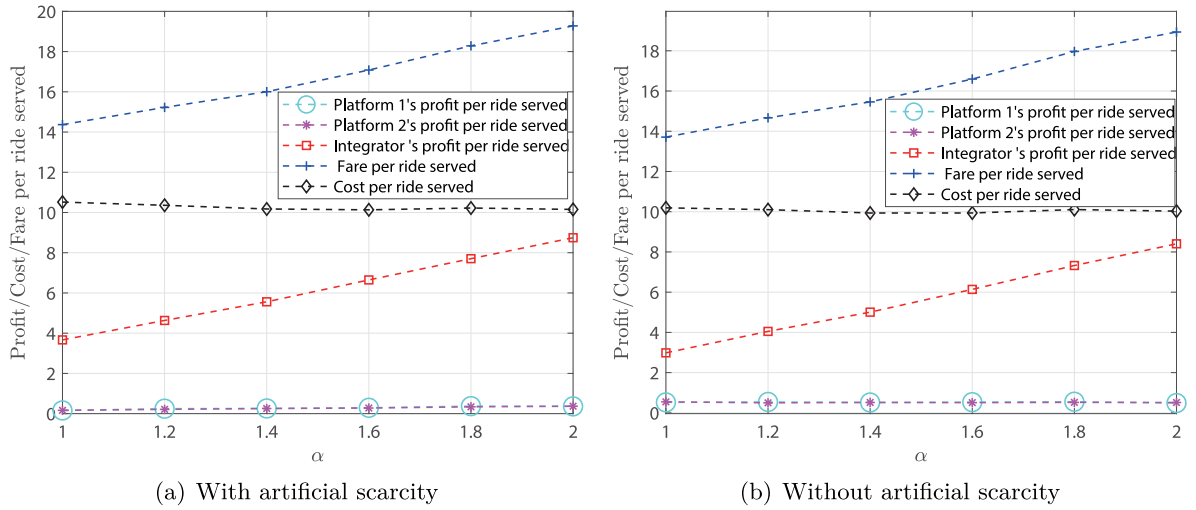


Fig. 4. Effect of  $\alpha$  (fare per unit distance) on profit/cost/fare per ride served.

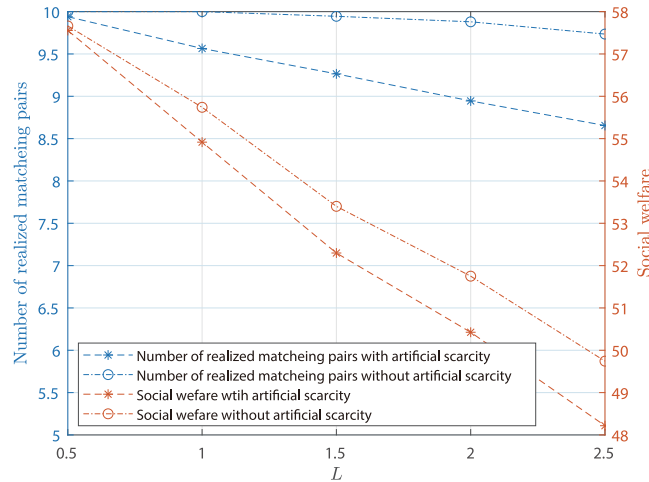


Fig. 5. Effect of  $L$  (side of the square region) on the number of matched pairs and social welfare. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

is almost negligible. We expect social welfare to be more sensitive to  $L$ , because lowering market density hurts social welfare by reducing not only the number of matched pairs, but also the profit margin for the pairs that do get matched. Indeed, as  $L$  increases five-fold, social welfare drops about 17% and 13%, respectively for the cases with and without artificial scarcity.

One can read from Fig. 6 that both the average cost and the average fare increase slowly with  $L$ , likely because the pickup phase constitutes a small portion of the entire delivery trip. Again, the integrator absorbs most profits with or without artificial scarcity. A noteworthy phenomenon concerns the effect of  $L$  on profitability. As the market becomes sparser (larger  $L$ ), the profitability of the integrator tends to fall, more so when artificial scarcity is forbidden. This can be explained by the squeeze on the supply due to the rising delivery cost, which leaves less room for the integrator to maneuver. Interestingly, a sparser market without artificial scarcity appears to favor the platforms, likely because a more tightly supplied market is a stronger position from which they can negotiate with the integrator. This observation motivates us to take a deeper dive into the effect of supply–demand imbalance in the next subsection.

### 6.2.3. Effect of supply–demand imbalance

We first vary the number of rides from 5 to 13, with an increment of 2. Fig. 7 plots the number of matched pairs and social welfare against the number of rides. As expected, more rides lead to more matched pairs, with or without artificial scarcity. However, since the supply is fixed at ten, the total number of matched pairs gradually approaches ten. As the number of rides increases from 5, the difference in productivity made by artificial scarcity first rises and peaks around 11, before falling back. It first rises because a severely oversupplied market (e.g., when  $|\mathcal{R}| = 5$ ) is less prone to the risk of artificial scarcity. The effect of artificial

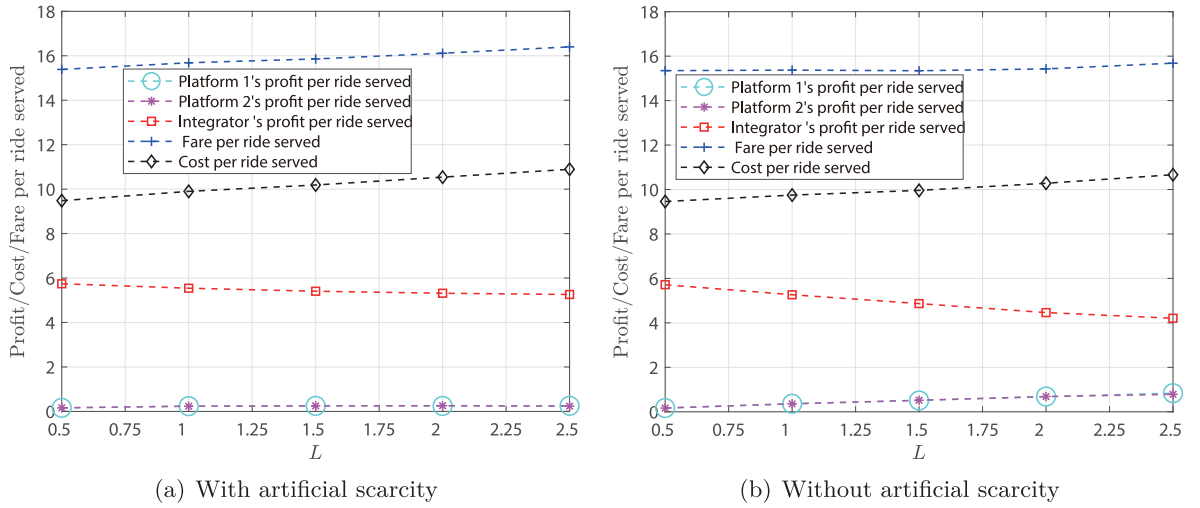


Fig. 6. Effect of  $L$  (side of the square region) on profit/cost/fare per ride served.

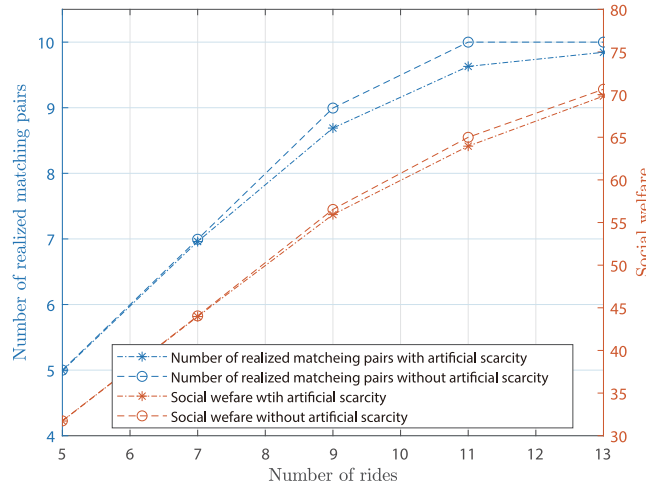


Fig. 7. Effect of supply-demand imbalance on the number of matched pairs and social welfare. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

scarcity on productivity drops once the market becomes severely under-supplied (e.g., when  $|\mathcal{R}| = 13$ ), because the likelihood of mismatching decreases as each vehicle is presented with more and more options. The pattern of social welfare tracks that of the matching productivity rather closely in this case.

Fig. 8 shows the average fare and cost gradually rise with the number of rides, possibly because a larger sample tends to increase the likelihood of drawing random values of trip length from the far end of the distribution, thereby pushing up the average of the sample. As long as the market is over-supplied ( $|\mathcal{R}| < 10$ ), the integrator still enjoys the overwhelming advantage in profitability over the platforms. The fortune is reversed only when (i) the market is under-supplied, and (ii) artificial scarcity is forbidden, see Fig. 8(b). Specifically, without artificial scarcity, the per capita profit of the two platforms begins to rise significantly when  $|\mathcal{R}| > 10$  and exceeds that of the integrator when  $|\mathcal{R}| = 13$ .

In a nutshell, the more the supply relative to demand, the less impactful artificial scarcity. In a tightly supplied market, the pressure to serve as many rides as possible — applied by the inclusion of the EAS constraints — forces the integrator to give up a considerable share of the profit that it would otherwise retain. With abundant supply, however, this pressure lets up. Unless the productivity is restricted by the supply side, eliminating artificial scarcity has limited effects on the system performance.

We next fix the fleet size of platform 1 and vary the fleet size of platform 2 from 2 to 10, with an increment of 2. When platform 2 has two vehicles, the total supply is seven, corresponding to an under-supplied market; if it has 10, the market is over-supplied with 15 vehicles for 10 rides. Fig. 9 plots the number of matched pairs against the fleet size of platform 2. We can see that rides served by each platform are approximately proportional to their fleet size. For example, when the fleet size of platform 2 is 10, it serves about 6.6 rides on average, which is exactly two thirds of the total rides served. Artificial scarcity plays a relatively minor

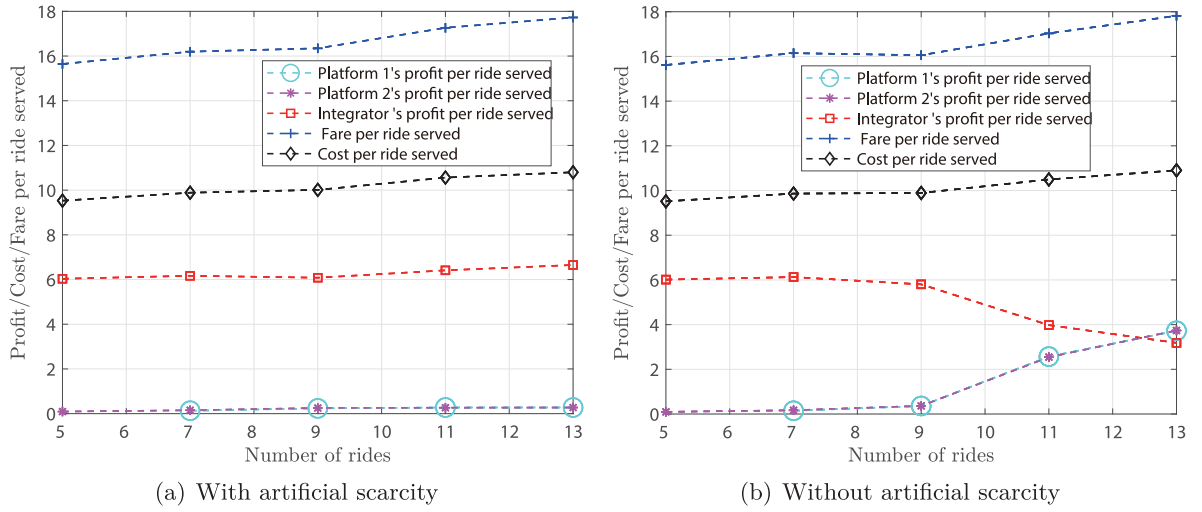


Fig. 8. Effect of supply-demand imbalance on profit/cost/fare per ride served.

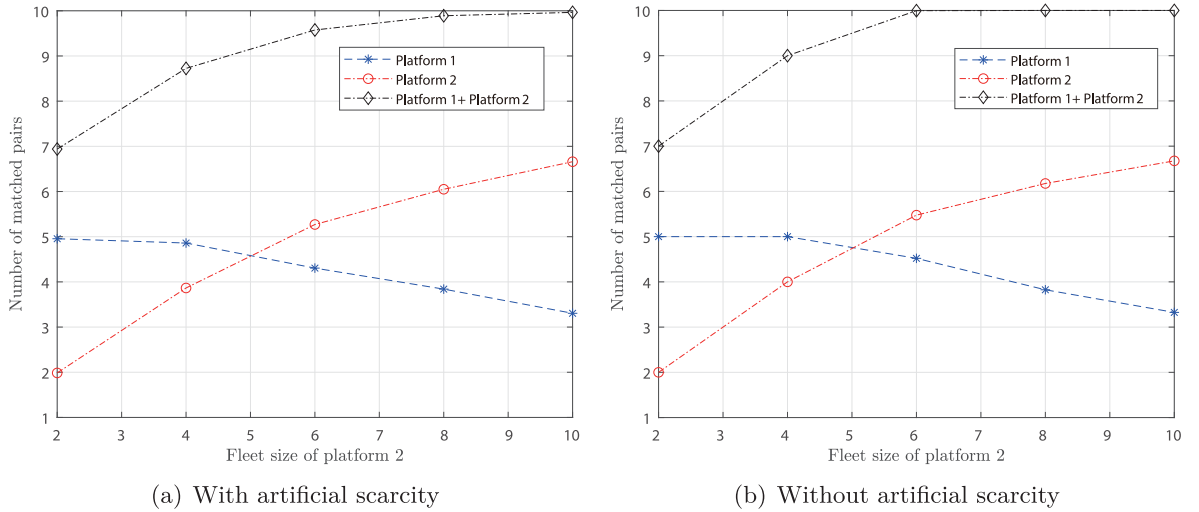


Fig. 9. Effect of the fleet size of platform 2 on the number of matched pairs.

role in shaping productivity, affecting the total number of matched pairs slightly and only when the fleet size of platform 2 ranges between 4 and 8.

Fig. 10 shows the average fare and cost per ride decrease with the size of platform 2's fleet, because having more vehicles in the same region increases the market density and thereby lowers the pickup time. The basic story is rather similar, although minor details differ. When artificial scarcity is allowed, most profits are taken by the integrator, regardless of the market conditions. When platform 2 has 4 or fewer vehicles (under-supplied market), eliminating artificial scarcity boosted the average profit of both platforms to a level comparable to that of the integrator. As platform 2 acquires more vehicles, however, the market gradually became over-supplied, eventually leading to the collapse of the profitability for both platforms that even the EAS constraint cannot prevent.

#### 6.2.4. Effect of reserve price

We fix the reserve price of platform 1 at  $\zeta_1 = 0.2$  and vary the reserve price of platform 2,  $\zeta_2$ , from 0.2 to 1, with an increment of 0.2. Fig. 11 plots the number of matched pairs against the reserve price of platform 2. We can see raising the reserve price charged by platform 2 generally reduces matching productivity for both platform 2 and the entire system, though the effect is more prominent with artificial scarcity. When artificial scarcity is permitted, the number of rides served by platform 2 decreases to zero when  $\zeta_2$  exceeds 0.8. At the same time, platform 1 enjoys a modest increase in the number of matched rides.

As shown in Fig. 12, social welfare decreases by about a third with or without artificial scarcity, when  $\zeta_2$  increases from 0.2 to 0.8. Interestingly, the trend is not monotone when artificial scarcity exists: social welfare reaches the lowest point when  $\zeta_2$  reaches

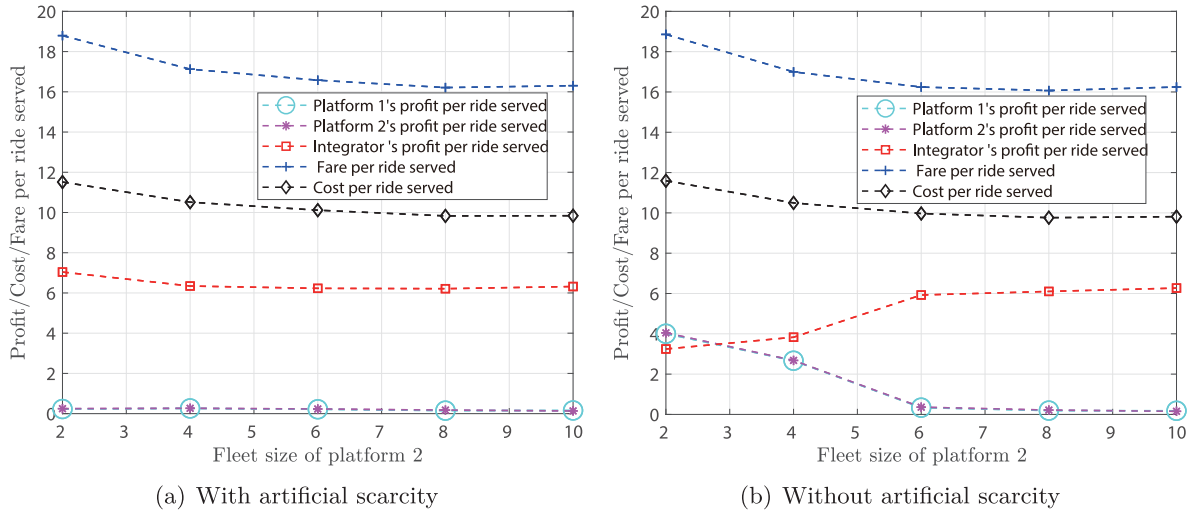


Fig. 10. Effect of the fleet size of platform 2 on profit/cost/fare per ride served.

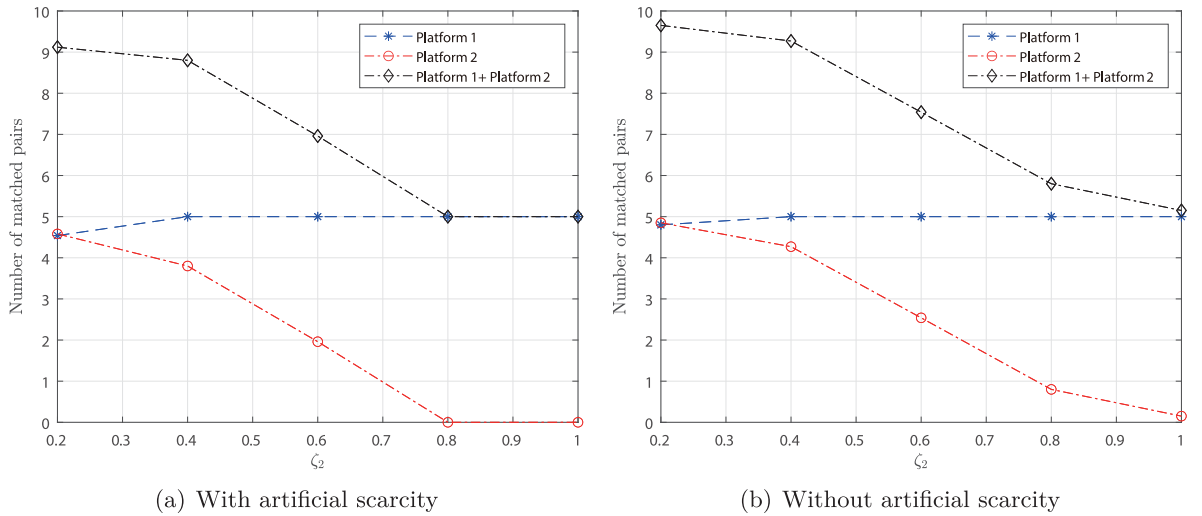


Fig. 11. Effect of reserve price of platform 2 on the number of matched pairs.

0.8 and begins to bounce back if the reserve price continues to rise. This is not surprising because, with artificial scarcity, social welfare is not always maximized.

Fig. 13 confirms the average fare and cost per ride served increase as platform 2 demands a higher reserve price. Eliminating artificial scarcity does not change the trend. The fortunes of the platforms and the integrator are affected by artificial scarcity very differently. With artificial scarcity (Fig. 13(a)), the per capita profit of the integrator first decreases with the reserve price, before rising again after  $\zeta_2$  exceeds a certain threshold. The trend is reversed for platforms 2. Notice that the number of matched pairs of platforms 2 is reduced to 0 when  $\zeta_2$  reaches 0.8 (hence the profit per ride curve stops abruptly at  $\zeta_2 = 0.6$  for platform 2). The per capita profit of platforms 1 basically stays the same. Once artificial scarcity is eliminated (Fig. 13(b)), however, the integrator's profitability declines and the platforms' profitability rises monotonically. Moreover, once  $\zeta_2$  exceeds 0.6, the integrator has a lower profitability than both platforms.

Fig. 14 compares the total profit earned by each player. The results highlight once again the critical role played by artificial scarcity. With artificial scarcity, the integrator is always the clear winner who grabs most profits. Without artificial scarcity, the share of the integrator's profit in the total profit decreases with the reserve price of platform 2. This benefits platform 1 greatly, because, as the supply on platform 2 becomes unusable due to the high reserve price, the overall supply in the market is tightened, which forces the integrator to share more profits with platform 1—this effect is closely related to what we have observed in Section 6.2.3.

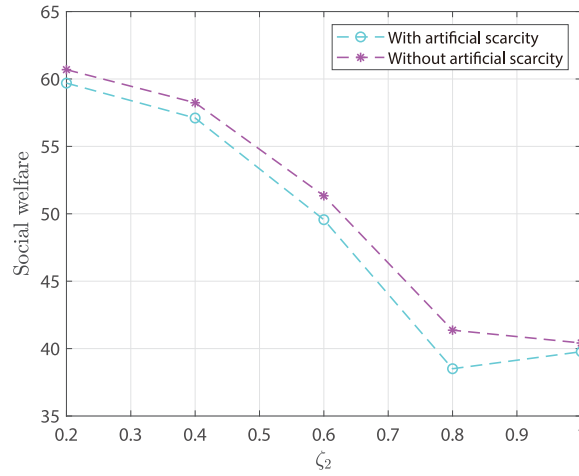


Fig. 12. Effect of reserve price of platform 2 on social welfare.

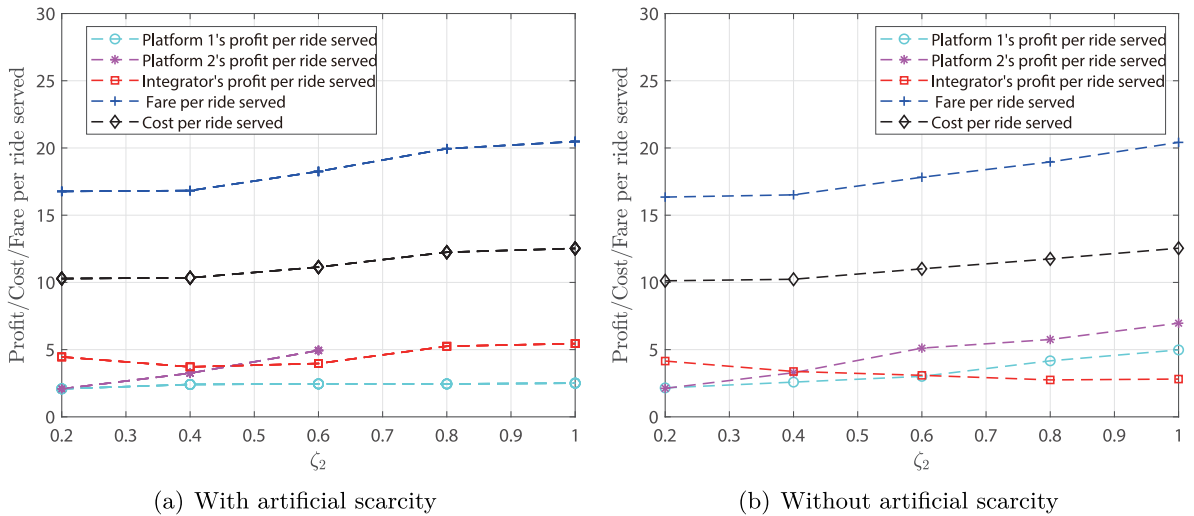


Fig. 13. Effect of the reserve price of platform 2 on profit/cost/fare per ride served.

## 7. Conclusions

We studied a ride-hail system in which a third-party integrator tries to allocate rides to a set of platforms in a manner that satisfies all stakeholders. Because the integrator has the monopoly power, the system is prone to artificial scarcity, which means it might intentionally price out the rides that some platforms would be willing to serve, in order to squeeze as much profit as possible. However, profit-maximization through artificial scarcity is myopic, because, in the long run, it can lower the overall productivity of the integrator, thus undermining the appeal to customers.

The ride allocation problem (RAP) can be viewed as a Stackelberg game. The integrator, as the leader, determines the profit-maximizing allocation by setting ride prices. In doing so, it ensures that no platform (follower) can find a better ride allocation at the given price. To eliminate artificial scarcity, a nonlinear constraint is imposed on the leader's optimization problem. We formulated RAP as a bilevel pricing program, which is then converted to a single-level optimization problem by dualizing the follower's problem. Since the nonlinear duality constraint has an equivalent linear form, the single-level problem can be turned into a mixed integer-linear program (MILP). We then showed this MILP equals its linear relaxation when all reserve price are zero, which effectively reduces an NP-hard problem to an "easy", polynomially solvable problem. Intrigued by this surprising result, we further investigated the relation between RAP and the cooperative assignment game, and confirmed they are indeed equivalent once artificial scarcity and reserve prices are stripped away. The main findings from our numerical experiments are summarized as follows.

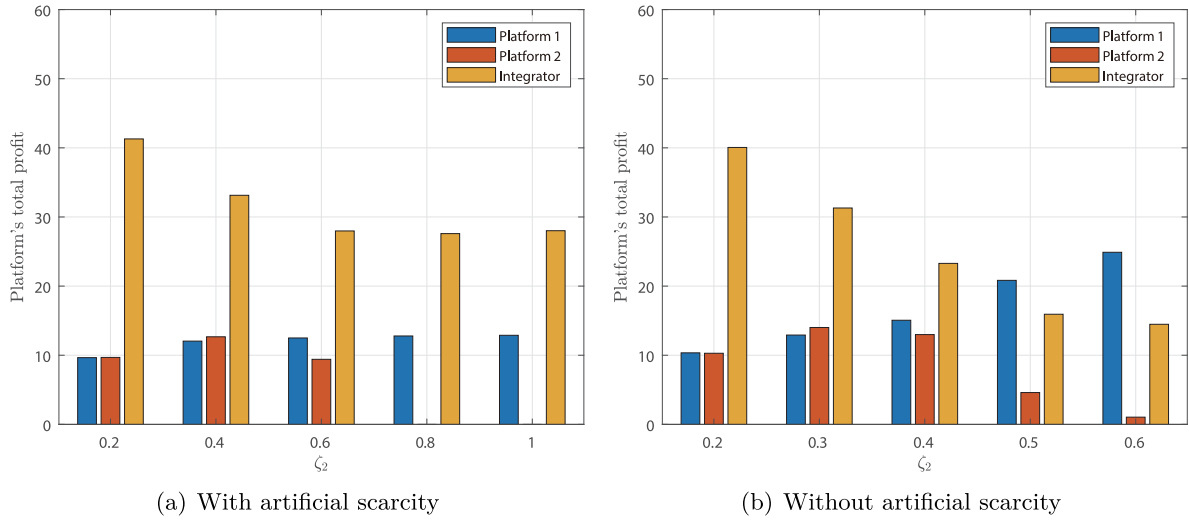


Fig. 14. Effect of the reserve profit of platform 2 on the total profit of the integrator and the platforms.

- Artificial scarcity negatively affects matching productivity (i.e., the number of rides served) and social welfare (i.e., the total profit). When reserve prices are not significant enough to alter the optimal allocation, the effect is relatively small, with the average reduction in the range of a few percent.
- The integrator is favored to take most profits, and leveraging artificial scarcity strengthens this dominant position. Moreover, the tighter the supply relative to the demand, the more benefits the integrator gets from artificial scarcity. On the other hand, the need to eliminate artificial scarcity forces the integrator to concede benefits to platforms. The tighter the supply, the more dramatic the change of the platforms' fortune.
- The reserve price is a double-edged sword for the platforms. In general, it helps redistribute benefits from the integrator to the platforms, but if overdone, can not only depress the platforms' profits but also undermine the matching productivity and social welfare of the system.

The present study treats the reserve price as given. In reality, the integrator might have to negotiate it with each platform. Unlike ride allocation, which takes place in real-time, this decision will be determined as a long-term contract. Consequently, a different game structure may be needed to properly model it. A future study can also consider the issue of incentive compatibility, i.e., how to ensure platforms report truthfully their delivery costs, which is also related to the negotiation of the reserve price. Another possibility is for the integrator to coordinate and centralize the calculation of each platform's delivery cost, which holds promise to further improve system performance by accounting for, among other things, the effect of dynamic supply–demand imbalance on the delivery cost. Finally, it would be interesting to explicitly model the competition between multiple integrators and independent platforms, which might reveal the mechanisms that incentivize them to willingly abandon artificial scarcity.

#### CRedit authorship contribution statement

**Ruijie Li:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Yang Liu:** Writing – review & editing, Supervision. **Xiaobo Liu:** Writing – review & editing. **Yu (Marco) Nie:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis.

#### Acknowledgments

The work was partially funded by Chinese National Science Foundation (CNSF) under the award number 72101217, by Sichuan Provincial Science and Technology Planning Project under the award number 2024NSFSC1053, by the Fundamental Research Funds for the Central University (Grant No. 2682023KJ009). The corresponding author's work was partially supported by the US National Science Foundation under the award number CMMI 1922665. The work of the first author and the second author was partially supported by Singapore Ministry of Education Academic Research Fund Tier 2 (MOE2019-T2-2-165).

#### Appendix A. Notations of main variables

See Table A.6.



**Table A.6**  
Notations of main variables.

| Symbol                                               | Descriptions                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{R}$                                        | Set of ride requests.                                                                                                                                                                                                                                                                                |
| $\mathcal{P}$                                        | Set of all real platforms.                                                                                                                                                                                                                                                                           |
| $\bar{p}$                                            | index of the virtual platform.                                                                                                                                                                                                                                                                       |
| $\bar{\mathcal{P}}$                                  | $\bar{\mathcal{P}} = \mathcal{P} \cup \{\bar{p}\}$ .                                                                                                                                                                                                                                                 |
| $\mathcal{V}_p$                                      | Set of available vehicles owned by platform $p \in \mathcal{P}$ .                                                                                                                                                                                                                                    |
| $\mathcal{V}$                                        | $\mathcal{V} = \bigcup_{p \in \mathcal{P}} \mathcal{V}_p$ .                                                                                                                                                                                                                                          |
| $\bar{\mathcal{V}}$                                  | $\bar{\mathcal{V}} = \mathcal{V} \cup \mathcal{V}_{\bar{p}}$ .                                                                                                                                                                                                                                       |
| $\bar{q}_r, \bar{q}$                                 | $\bar{q}_r$ is the fare paid by rider $r$ to the integrator, $\bar{q} = \{\bar{q}_r, r \in \mathcal{R}\}$ .                                                                                                                                                                                          |
| $q_r, q$                                             | Price platforms must accept if they are to serve ride $r$ without (with) the virtual platform, $q = \{q_r\}$ , where $r \in \mathcal{R}$ .                                                                                                                                                           |
| $\bar{q}_r, \bar{q}$                                 | Price platforms must accept if they are to serve ride $r$ with the virtual platform, $\bar{q} = \{\bar{q}_r\}$ , where $r \in \mathcal{R}$ .                                                                                                                                                         |
| $c_{vr}, c$                                          | Cost required to deliver ride $r$ by vehicle $v$ without virtual platform, $c = \{c_{vr}\}$ , where $r \in \mathcal{R}, v \in \mathcal{V}$ .                                                                                                                                                         |
| $\bar{c}_{vr}, \bar{c}$                              | Cost required to deliver ride $r$ by vehicle $v$ with virtual platform, $\bar{c} = \{\bar{c}_{vr}\}$ , where $r \in \mathcal{R}, v \in \mathcal{V}$ .                                                                                                                                                |
| $\zeta_p$                                            | Reserve profit of platform $p$ .                                                                                                                                                                                                                                                                     |
| $x_{vr}, \mathbf{x}_p, \mathbf{x}$                   | Allocation decision without virtual platform. $x_{vr} = 1$ if the matching pair $(v, r)$ is realized; and 0 otherwise. $\mathbf{x}_p = \{x_{vr}\}$ , where $r \in \mathcal{R}, v \in \mathcal{V}_p$ ; $\mathbf{x} = \{\mathbf{x}_p\}$ , where $p \in \mathcal{P}$ .                                  |
| $\bar{x}_{vr}, \bar{\mathbf{x}}_p, \bar{\mathbf{x}}$ | Allocation decision with virtual platform. $\bar{x}_{vr} = 1$ if the matching pair $(v, r)$ is realized; and 0 otherwise. $\bar{\mathbf{x}}_p = \{\bar{x}_{vr}\}$ , where $r \in \mathcal{R}, v \in \mathcal{V}_p$ ; $\bar{\mathbf{x}} = \{\bar{\mathbf{x}}_p\}$ , where $p \in \bar{\mathcal{P}}$ . |
| $\kappa_{rp}, \kappa_p, \kappa$                      | $\kappa_{rp}$ is the dual variables corresponding to constraints (10b). $\kappa_p = \{\kappa_{rp}\}$ , where $r \in \mathcal{R}$ and $v \in \mathcal{V}_p$ ; $\kappa = \{\kappa_p\}$ , where $p \in \bar{\mathcal{P}}$ .                                                                             |
| $\varphi_v, \varphi_p, \varphi$                      | $\varphi_v$ is the dual variables corresponding to constraints (10c). $\varphi_p = \{\varphi_v\}$ , where $r \in \mathcal{R}$ and $v \in \mathcal{V}_p$ ; $\varphi = \{\varphi_p\}$ , where $p \in \bar{\mathcal{P}}$ .                                                                              |
| $\delta_{vr}, \delta_p, \delta$                      | $\delta_{vr}$ is the dual variables corresponding to constraints (13). $\delta_p = \{\delta_{vr}\}$ , where $r \in \mathcal{R}$ and $v \in \mathcal{V}_p$ ; $\delta = \{\delta_p\}$ , where $p \in \bar{\mathcal{P}}$ .                                                                              |

## Appendix B. Proof of Proposition 2

To prove (i), we first note  $\mathbf{x}^*$  satisfies the standard constraints ((5b)–(5c) and (6b)–(6c)) from the bipartite matching problem since  $\mathbf{x}^*$  is a subset of  $\bar{\mathbf{x}}^*$ .  $[\mathbf{x}^*, \mathbf{q}^*]$  must also satisfy the EAS constraint since Constraints (9d) and (5d) are identical. Similarly, Constraint (10d) is identical to Constraint (6d).

To prove (ii), we only need to show the following: if  $[\mathbf{x}^*, \mathbf{q}^*]$  is not an optimal solution to the bilevel program (5)–(6), then neither is  $[\bar{\mathbf{x}}^*, \bar{\mathbf{q}}^*]$  to the bilevel program (9)–(10).

We first note the virtual platform makes no money by construction, which means the integrator cannot make any money from rides assigned to the virtual platform. Thus, by stripping away the matched pairs associated with the virtual platform, the profits of both real platforms and the integrator will remain intact. It follows

$$L(\mathbf{x}^*, \mathbf{q}^*) = L(\bar{\mathbf{x}}^*, \bar{\mathbf{q}}^*). \quad (25)$$

Since we assume  $[\mathbf{x}^*, \mathbf{q}^*]$  is not an optimal solution to the bilevel program (5)–(6), let  $[\mathbf{x}^{**}, \mathbf{q}^{**}]$  be one. It follows

$$L(\mathbf{x}^{**}, \mathbf{q}^{**}) > L(\bar{\mathbf{x}}^*, \bar{\mathbf{q}}^*). \quad (26)$$

Based on  $[\mathbf{x}^{**}, \mathbf{q}^{**}]$ , we can construct a feasible solution  $[\bar{\mathbf{x}}^{**}, \bar{\mathbf{q}}^{**}]$  by leaving everything untouched except assigning the unmatched rides to the virtual platform. This would not change the integrator's profit, however. Therefore, we have

$$L(\bar{\mathbf{x}}^{**}, \bar{\mathbf{q}}^{**}) = L(\mathbf{x}^{**}, \mathbf{q}^{**}). \quad (27)$$

Combining Eqs. (25), (26), and (27) leads to

$$L(\bar{\mathbf{x}}^{**}, \bar{\mathbf{q}}^{**}) > L(\bar{\mathbf{x}}^*, \bar{\mathbf{q}}^*), \quad (28)$$

which suggests  $[\bar{\mathbf{x}}^*, \bar{\mathbf{q}}^*]$  is not an optimal solution to the bilevel program (9)–(10), a contradiction. This completes the proof.

## Appendix C. Proof of Proposition 3

Let  $[\bar{\mathbf{x}}^*, \bar{\mathbf{q}}^*]$  be an optimal solution to the bilevel program (9)–(10). Any ride  $r$  must fall into one of the two categories.

- Case 1: ride  $r$  is matched with a real vehicle, i.e.,  $\sum_{v \in \mathcal{V}} \bar{x}_{vr}^* = 1$ .
- Case 2: ride  $r$  is not matched with any real vehicle, i.e.,  $\sum_{v \in \mathcal{V}} \bar{x}_{vr}^* = 0$ .

The first case obviously satisfies Constraint (9d). For the second case, we need to show  $\bar{q}_r^* = \bar{q}_r$ . Recall in the transformed problem  $r$  is assigned to vehicle  $v' \in \mathcal{V}_{\bar{p}}$ . We have  $\bar{c}_{v'r} = \bar{q}_r$  by construction and  $\bar{q}_r^* \leq \bar{q}_r = \bar{c}_{v'r}$  to ensure the assignment does not yield a loss for the integrator. Because the virtual platform cannot lose money ( $\bar{c}_{v'r} \leq \bar{q}_r^*$ ), we should have  $\bar{q}_r^* = \bar{c}_{v'r} = \bar{q}_r$ . This completes the proof.

#### Appendix D. Proof of Proposition 4

Necessity. We first show if Constraint (16) is satisfied for every  $p$ , then Constraint (18) must be satisfied.

$$\begin{aligned} \sum_{p \in \bar{\mathcal{P}}} \left( \sum_{r \in \mathcal{R}} \kappa_{rp} + \sum_{v \in \mathcal{V}_p} \varphi_v \right) &= \sum_{p \in \bar{\mathcal{P}}} \left( \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_p} (\bar{q}_r - \bar{c}_{vr}) \bar{x}_{vr} \right) \\ &= \sum_{r \in \mathcal{R}} \bar{q}_r \left( \sum_{p \in \bar{\mathcal{P}}} \sum_{v \in \mathcal{V}_p} \bar{x}_{vr} \right) - \sum_{r \in \mathcal{R}} \sum_{p \in \bar{\mathcal{P}}} \sum_{v \in \mathcal{V}_p} \bar{c}_{vr} \bar{x}_{vr} \\ &= \sum_{r \in \mathcal{R}} \bar{q}_r - \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}} \bar{c}_{vr} \bar{x}_{vr}. \end{aligned}$$

The third equality holds because of constraint (9b) (i.e., all ride requests must be matched after the virtual platform is added).

Sufficiency: We next prove condition (18) leads to condition (16) by reduction to absurdity. Suppose condition (18) holds but there exists  $p_1 \in \bar{\mathcal{P}}$  such that condition (16) is violated, i.e.,

$$\sum_{r \in \mathcal{R}} \kappa_{rp_1} + \sum_{v \in \mathcal{V}_{p_1}} \varphi_v \neq \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_{p_1}} (\bar{q}_r - \bar{c}_{vr}) \bar{x}_{vr}, p_1 \in \bar{\mathcal{P}}. \quad (29)$$

The weak duality theorem requires

$$\sum_{r \in \mathcal{R}} \kappa_{rp_1} + \sum_{v \in \mathcal{V}_{p_1}} \varphi_v > \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_{p_1}} (\bar{q}_r - \bar{c}_{vr}) \bar{x}_{vr}. \quad (30)$$

Combining (30) and (18), we can see there must exist  $p_2 \in \bar{\mathcal{P}}$  such that

$$\sum_{r \in \mathcal{R}} \kappa_{rp_2} + \sum_{v \in \mathcal{V}_{p_2}} \varphi_v < \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_{p_2}} (\bar{q}_r - \bar{c}_{vr}) \bar{x}_{vr}. \quad (31)$$

However, this is impossible per the weak duality theorem.

#### Appendix E. Proof of Theorem 1

The optimal solution of (21) must be an upper bound of (20). Thus, we only need to show that, for any feasible solution  $[\bar{x}1, \bar{q}1, \kappa, \varphi]$  to Problem (21), there must exist an integral solution  $[\bar{x}2, \bar{q}2, \kappa, \varphi]$  that corresponds to an equal or greater profit. With  $[\kappa, \varphi]$  fixed, the objective (21a) equals

$$\min_{\bar{x}, \bar{q}} \sum_{r \in \mathcal{R}} \bar{q}_r \quad (32)$$

since  $\sum_{r \in \mathcal{R}} \bar{q}_r$  is a constant. Recalling Eq. (18), (32) can be further converted to

$$\min_{\bar{x}} \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}} \bar{c}_{vr} \bar{x}_{vr}. \quad (33)$$

Given  $[\bar{x}1, \bar{q}1, \kappa, \varphi]$ , we construct a new feasible solution  $[\bar{x}2, \bar{q}2]$  as follows.

- Consider the bipartite matching problem

$$\min_{\bar{x}} \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}} \bar{c}_{vr} \bar{x}_{vr} \quad (34a)$$

subject to:

$$(9b), (9c), (9e). \quad (34b)$$

We can always find an integral optimal solution  $\bar{x}2$  to Problem (34) since its coefficient matrix is totally unimodular.

- Set  $\bar{q}2$  such that

$$\bar{q}2_1 = \bar{q}1_1 - \left( \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}} \bar{c}_{vr} (\bar{x}1_{vr} - \bar{x}2_{vr}) \right), \quad (35a)$$

$$\bar{q}2_r = \bar{q}1_r, \forall r \neq 1. \quad (35b)$$

We proceed to show  $[\bar{x}2, \bar{q}2, \kappa, \varphi]$  is a feasible solution to Problem (20). First, Constraints (9b) and (9c) in Problem (20) are satisfied since  $\bar{x}2$  solves (34), which includes the same constraints. Second, as  $\bar{x}2$  is an optimal solution to Problem (34), it follows that

$$\sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}} \bar{c}_{vr} \bar{x}2_{vr} \leq \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}} \bar{c}_{vr} \bar{x}1_{vr}. \quad (36)$$

Combining (35) and (36), we have

$$\bar{q}2_r \leq \bar{q}1_r, \forall r \in \mathcal{R}. \quad (37)$$

Because  $[\bar{x}1, \bar{q}1, \kappa, \varphi]$  satisfies Constraint (20d), we must have

$$\kappa_{rp} + \varphi_v \geq \bar{q}1_r - \bar{c}_{vr}, \rightarrow \kappa_{rp} + \varphi_v \geq \bar{q}2_r - \bar{c}_{vr}, \forall r \in \mathcal{R}, \forall v \in \bar{\mathcal{V}}, \forall p \in \bar{\mathcal{P}}. \quad (38)$$

Therefore, (20d) will be satisfied by  $[\bar{x}2, \bar{q}2, \kappa, \varphi]$ . Note that the non-negativity constraint (14c) is not affected by the change. Third,  $[\bar{x}2, \bar{q}2, \kappa, \varphi]$  must also satisfy Constraint (18) because

$$\begin{aligned} \sum_{p \in \bar{\mathcal{P}}} \left( \sum_{r \in \mathcal{R}} \kappa_{rp} + \sum_{v \in \bar{\mathcal{V}}} \varphi_v \right) &= \sum_{r \in \mathcal{R}} \bar{q}1_r - \sum_{r \in \mathcal{R}} \sum_{v \in \bar{\mathcal{V}}} \bar{c}_{vr} \bar{x}1_{vr} \\ &= \sum_{r \in \mathcal{R}} \bar{q}2_r + \left( \sum_{r \in \mathcal{R}} \sum_{v \in \bar{\mathcal{V}}} \bar{c}_{vr} (\bar{x}1_{vr} - \bar{x}2_{vr}) \right) - \sum_{r \in \mathcal{R}} \sum_{v \in \bar{\mathcal{V}}} \bar{c}_{vr} \bar{x}1_{vr} \\ &= \sum_{r \in \mathcal{R}} \bar{q}2_r - \sum_{r \in \mathcal{R}} \sum_{v \in \bar{\mathcal{V}}} \bar{c}_{vr} \bar{x}2_{vr}. \end{aligned} \quad (39)$$

The second equality above holds because of (35).

Finally, we show  $[\bar{x}2, \bar{q}2, \kappa, \varphi]$  must produce an equal or greater profit compared to  $[\bar{q}1, \bar{x}1, \kappa, \varphi]$ . To see this, note Eq. (37) implies

$$\sum_{r \in \mathcal{R}} \bar{q}2_r \leq \sum_{r \in \mathcal{R}} \bar{q}1_r, \quad (40)$$

which, recalling (32), indicates  $[\bar{x}2, \bar{q}2, \kappa, \varphi]$  corresponds to an equal or greater profit.

## Appendix F. Proof of Proposition 5

We only need to show that, for any feasible solution  $[\eta, \kappa, \varphi]$  to Problem (22), there must exist a feasible solution  $[\bar{\eta}, 0, \varphi]$  corresponding to an equal or greater objective function value. We note  $[\bar{\eta}, 0, \varphi]$  is a feasible solution to Problem (22), because it satisfies Constraint (22b) by construction and Constraint (20d) due to

$$\bar{\eta}_r + \varphi_v = \eta_r + \sum_{p \in \bar{\mathcal{P}}} \kappa_{rp} + \varphi_v \geq \eta_r + \kappa_{rp} + \varphi_v \geq \bar{q}_r - \bar{c}_{vr}. \quad (41)$$

Moreover,  $[\bar{\eta}, 0, \varphi]$  give an equal or greater objective function value since

$$\sum_{r \in \mathcal{R}} \bar{\eta}_r = \sum_{r \in \mathcal{R}} \eta_r + \sum_{r \in \mathcal{R}} \sum_{p \in \bar{\mathcal{P}}} \kappa_{rp} \geq \sum_{r \in \mathcal{R}} \eta_r. \quad (42)$$

This completes the proof.

## References

- Agatz, Niels, Erera, Alan, Savelsbergh, Martin, Wang, Xing, 2012. Optimization for dynamic ride-sharing: A review. *European J. Oper. Res.* 223 (2), 295–303.
- Alonso-Mora, Javier, Samaranayake, Samitha, Wallar, Alex, Frazzoli, Emilio, Rus, Daniela, 2017. On-demand high-capacity ride-sharing via dynamic trip-vehicle assignment. *Proc. Natl. Acad. Sci.* 114 (3), 462–467.
- Bao, Yue, Zang, Guangzhi, Yang, Hai, Gao, Ziyou, Long, Jiancheng, 2023. Mathematical modeling of the platform assignment problem in a ride-sourcing market with a third-party integrator. *Transp. Res. B* 178, 102833.
- Bialas, Wayne F., Karwan, Mark H., 1984. Two-level linear programming. *Manag. Sci.* 30 (8), 1004–1020.
- Bronfenbrenner, Martin, 1962. The scarcity hypothesis in modern economics. *Am. J. Econ. Sociol.* 21 (3), 265–270.
- Brotcorne, Luce, Labbé, Martine, Marcotte, Patrice, Savard, Gilles, 2001. A bilevel model for toll optimization on a multicommodity transportation network. *Transp. Sci.* 35 (4), 345–358.
- Castillo, Juan Camilo, Knoepfle, Dan, Weyl, Glen, 2017. Surge pricing solves the wild goose chase. In: *Proceedings of the 2017 ACM Conference on Economics and Computation*. pp. 241–242.
- Chen, Xiqun Michael, Zahiri, Majid, Zhang, Shuaichao, 2017. Understanding ridesplitting behavior of on-demand ride services: An ensemble learning approach. *Transp. Res. C* 76, 51–70.
- Colson, Benoît, Marcotte, Patrice, Savard, Gilles, 2007. An overview of bilevel optimization. *Ann. Oper. Res.* 153 (1), 235–256.
- Cramer, Judd, Krueger, Alan B., 2016. Disruptive change in the taxi business: The case of uber. *Amer. Econ. Rev.* 106 (5), 177–182.
- Demange, Gabrielle, 1982. Strategyproofness in the Assignment Market Game. *Laboratoire d'économétrie de l'École polytechnique*.
- Dempe, Stephan, 2019. Computing locally optimal solutions of the bilevel optimization problem using the kkt approach. In: *International Conference on Mathematical Optimization Theory and Operations Research*. Springer, pp. 147–157.
- Ding, Xiaoshu, Qi, Qi, Jian, Sisi, Yang, Hai, 2023. Mechanism design for mobility-as-a-service platform considering travelers' strategic behavior and multidimensional requirements. *Transp. Res. B* 173, 1–30.
- Dong, Tingting, Sun, Xiaotong, Luo, Qi, Wang, Jian, Yin, Yafeng, 2024. The dual effects of team contest design on on-demand service work schedules. *Serv. Sci.* 16 (1), 22–41.
- Farvareh, Hamid, Sepehri, Mohammad Mehdi, 2011. A single-level mixed integer linear formulation for a bi-level discrete network design problem. *Transp. Res. E* 47 (5), 623–640.
- Fontaine, Pirmin, Minner, Stefan, 2014. Benders decomposition for discrete-continuous linear bilevel problems with application to traffic network design. *Transp. Res. B* 70, 163–172.
- Furuhata, Masabumi, Dessouky, Maged, Ordóñez, Fernando, Brunet, Marc Etienne, Wang, Xiaoqing, Koenig, Sven, 2013. Ridesharing: The state-of-the-art and future directions. *Transp. Res. B* 57 (57), 28–46.
- Gale, David, Shapley, Lloyd S., 1962. College admissions and the stability of marriage. *Amer. Math. Monthly* 69 (1), 9–15.

- Grimm, Veronika, Orlinskaya, Galina, Schewe, Lars, Schmidt, Martin, Zöttl, Gregor, 2021. Optimal design of retailer-prosumer electricity tariffs using bilevel optimization. *Omega* 102, 102327.
- Kleinert, Thomas, Labbé, Martine, Ljubić, Ivana, Schmidt, Martin, 2021. A survey on mixed-integer programming techniques in bilevel optimization. *EURO J. Comput. Optim.* 9, 100007.
- Kleinert, Thomas, Schmidt, Martin, 2021. Computing feasible points of bilevel problems with a penalty alternating direction method. *INFORMS J. Comput.* 33 (1), 198–215.
- Labbé, Martine, Marcotte, Patrice, Savard, Gilles, 1998. A bilevel model of taxation and its application to optimal highway pricing. *Manag. Sci.* 44 (12-part-1), 1608–1622.
- Mitropoulos, Lambros, Kortsari, Annie, Ayfantopoulou, Georgia, 2021. A systematic literature review of ride-sharing platforms, user factors and barriers. *Eur. Transp. Res. Rev.* 13 (1), 1–22.
- Mourad, Abood, Puchinger, Jakob, Chu, Chengbin, 2019. A survey of models and algorithms for optimizing shared mobility. *Transp. Res. B* 123, 323–346.
- Nie, Yu Marco, 2017. How can the taxi industry survive the tide of ridesourcing ? Evidence from Shenzhen, China. *Transp. Res. C* 79, 242–256.
- Pandey, Venkatesh, Monteil, Julien, Gambella, Claudio, Simonetto, Andrea, 2019. On the needs for maas platforms to handle competition in ridesharing mobility. *Transp. Res. C* 108, 269–288.
- Pantelidis, Theodoros P., Chow, Joseph Y.J., Rasulkhani, Saeid, 2020. A many-to-many assignment game and stable outcome algorithm to evaluate collaborative mobility-as-a-service platforms. *Transp. Res. B* 140, 79–100.
- Qin, Zhiwei Tony, Zhu, Hongtu, Ye, Jieping, 2021. Reinforcement learning for ridesharing: A survey. In: 2021 IEEE International Intelligent Transportation Systems Conference. ITSC, IEEE, pp. 2447–2454.
- Rasulkhani, Saeid, Chow, Joseph Y.J., 2019. Route-cost-assignment with joint user and operator behavior as a many-to-one stable matching assignment game. *Transp. Res. B* 124, 60–81.
- Shapley, Lloyd S., Shubik, Martin, 1969. On the core of an economic system with externalities. *Am. Econ. Rev.* 59 (4), 678–684.
- Shapley, Lloyd S., Shubik, Martin, 1971. The assignment game I: The core. *Int. J. Game Theory* 1 (1), 111–130.
- Sotomayor, Marilda, 1999. The lattice structure of the set of stable outcomes of the multiple partners assignment game. *Int. J. Game Theory* 28 (4), 567–583.
- Tafreshian, Amirmahdi, Masoud, Neda, Yin, Yafeng, 2020. Frontiers in service science: Ride matching for peer-to-peer ride sharing: A review and future directions. *Serv. Sci.* 12 (2–3), 44–60.
- Tamir, Arie, 1976. On totally unimodular matrices. *Networks* 6 (4), 373–382.
- Wang, Hai, Yang, Hai, 2019. Ridesourcing systems: A framework and review. *Transp. Res. B* 129, 122–155.
- Xi, Haoning, Aussel, Didier, Liu, Wei, Waller, S Travis, Rey, David, 2022. Single-leader multi-follower games for the regulation of two-sided mobility-as-a-service markets. *European J. Oper. Res.* 317 (3), 718–736.
- Zhang, Kenan, Nie, Yu Marco, 2021. Inter-platform competition in a regulated ride-hail market with pooling. *Transp. Res. E* 151, 102327.
- Zhou, Yaqian, Yang, Hai, Ke, Jintao, Wang, Hai, Li, Xinwei, 2022. Competition and third-party platform-integration in ride-sourcing markets. *Transp. Res. B* 159, 76–103.
- Zugno, Marco, Morales, Juan Miguel, Pinson, Pierre, Madsen, Henrik, 2013. A bilevel model for electricity retailers' participation in a demand response market environment. *Energy Econ.* 36, 182–197.